

CORE MATHEMATICS GRADE 10: QUADRATIC EXPRESSIONS AND EQUATIONS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.3 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Core Mathematics
Strand	Strand 1.0: Numbers and Algebra
Sub-Strand	Sub-Strand 1.3: Quadratic Expressions and Equations
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Formation of quadratic expressions – Quadratic identities derived from the concept of area: $(a + b)^2$, $(a - b)^2$, $a^2 - b^2$ – Factorisation of quadratic expressions – Formation of quadratic equations from real-life situations – Solving quadratic equations by factorisation – Application of quadratic equations in real-life situations
Learning Outcomes	NOTE FOR REVIEWER: The OCR text supplied did not include the Sub-Strand 1.3 outcomes page. The outcomes below are reconstructed from the KICD assessment-rubric indicators for this sub-strand (form quadratic expressions; derive quadratic identities from the concept of area; factorise quadratic expressions; form, solve and apply quadratic equations in real-life situations). Check the wording against the source PDF before release. By the end of the sub-strand, the learner should be able to: - form quadratic expressions - derive quadratic identities from the concept of area - factorise quadratic expressions - form quadratic equations from real-life situations - solve quadratic equations - apply quadratic equations in real-life situations
Core Competencies	NOTE: Not in the supplied OCR text; verify against the KICD page. – Critical thinking and problem solving: learners form and solve quadratic equations from the fencing and plot-area situations and test whether each solution is sensible. – Communication and collaboration: learners discuss area models and factorisation strategies in groups and justify their steps to peers. – Learning to learn: learners revise their cumulative model of quadratic relationships after each lesson. – Creativity and imagination: learners design plots, paths and frames using locally available materials.
Core Values	NOTE: Not in the supplied OCR text; verify against the KICD page. – Responsibility: learners handle measuring tools and cut-out materials with care and complete group tasks.

	– Integrity: learners measure and record areas accurately and check solutions honestly. – Unity: learners work together in teams to build area models and share solution methods. – Respect: learners listen to the reasoning of others when strategies differ.
Science & Engineering Practices	– Asking questions and defining problems: learners generate questions about why two different plot widths give the same area. – Developing and using models: learners use area tiles and algebra-tile models to represent quadratic expressions and identities. – Analyzing and interpreting data: learners tabulate width against area for a fixed fence length and interpret the pattern. – Using mathematics and computational thinking: learners form, factorise and solve quadratic equations. – Constructing explanations: learners explain why a quadratic equation can have two solutions and why one may be rejected. – Engaging in argument from evidence: learners defend their choice of plot dimensions using calculations.
Pertinent & Contemporary Issues (PCIs)	NOTE: Not in the supplied OCR text; verify against the KICD page. – Education for Sustainable Development: learners plan efficient use of farm land, fencing and materials. – Life skills (decision making): learners choose plot dimensions using mathematical evidence. – Financial literacy: learners compute fencing and seed costs for plots in shillings.
Career Connections	– Land surveyor: calculating plot areas and boundaries. – Civil engineer / architect: designing floor areas and structures. – Agricultural officer / farmer: planning plot layout for maize, beans and sukuma wiki. – Quantity surveyor: costing materials such as fencing, tiles and paving. – Software developer / data analyst: modelling relationships with quadratic functions.
Focus for Lessons	This unit uses a Kenyan farm-plot situation to show learners that quadratic expressions arise naturally from area. Learners move from area models to identities, factorisation and solving equations, then apply these to real contexts. Each lesson follows Predict, Observe, Explain, DQB and Model Building, so learners revise one cumulative model of quadratic relationships.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How can we use quadratic expressions and equations to find the dimensions of a plot when its fence length and area are both fixed, and why do two different plots sometimes have the same area? KICD KEY INQUIRY QUESTIONS (not included in the supplied OCR text; verify wording against the KICD page): 1. How do we form quadratic expressions and equations? 2. Where do we apply quadratic equations in real-life situations?
Anchoring Phenomenon	A farmer in Kericho has exactly 40 m of fencing wire for a rectangular vegetable plot (sukuma wiki and spinach) beside her house. When her children try different widths, they notice that a plot 8 m wide and a plot 12 m wide both enclose exactly 96 m². A plot 10 m wide gives a larger area of 100 m², and any width above 20 m is impossible. Learners ask: with the same fence, why do two different shapes give the same area, why is there a maximum, and how can we find the widths for a target area without trial and error? Learners observe the pattern with string loops, grid paper and tile models and find that it is described by an expression with a squared term, a quadratic.
Storyline Thread	Lesson 1: Forming quadratic expressions. Learners predict how area changes with width for a 40 m fence, then use grid paper to build width × length expressions such as $x(20 - x)$ and $(x + 3)(x + 5)$. DQB: Why does the area follow a pattern with x^2? Lesson 2: Deriving identities from area. Learners use area tiles and cut-out squares to derive $(a + b)^2 = a^2 + 2ab + b^2$ and $(a - b)^2 = a^2 - 2ab + b^2$, and test them with plots of 10 m, 12 m and 8 m. Model v1: area models for expansions. Lesson 3: Difference of two squares and applying identities. Learners cut a small square from a large square, rearrange the pieces

	to derive $a^2 - b^2 = (a - b)(a + b)$, and use it for quick mental calculations such as $52^2 - 48^2$. Model revised with all three identities. Lesson 4: Factorising $x^2 + bx + c$. Learners predict the dimensions of a rectangle from a given area expression, rebuild it with tiles, and factorise by finding two numbers with a given sum and product. Link back to the 8 m and 12 m plots. Lesson 5: Factorising $ax^2 + bx + c$ and special cases. Learners use the grouping method and the area-tile method to factorise expressions such as $2x^2 + 7x + 3$, and check by expansion. DQB review: which methods work best, and when? Lesson 6: Solving quadratic equations by factorisation. Learners use the zero-product idea to solve $x^2 - 20x + 96 = 0$ and explain why $x = 8$ and $x = 12$ both appear, resolving the central puzzle of the phenomenon. Model revised to include two solutions. Lesson 7: Forming quadratic equations from real-life situations. Learners translate the mandazi tray, path-around-shamba and thrown-ball situations into equations and solve them, rejecting solutions that make no sense, such as negative lengths. Lesson 8: Applying and synthesising. Learners use their full model to design a school-garden plot of a target area with a given fence length, justify their dimensions to the class, and explain the maximum area at 10 m. They finalise the DQB and the cumulative model.
Supporting Phenomena	– A path of uniform width around a rectangular maize shamba in Nakuru changes the total area in a way that is not simply proportional to the path width. – A paving contractor in Kisumu notices that a square courtyard with a square cut out has area equal to a product of two sums, $a^2 - b^2 = (a - b)(a + b)$. – A mandazi vendor in Mombasa wants a rectangular tray whose length is 3 cm more than its width and whose area is 180 cm². – A ball thrown by a school athlete at Kasarani passes the same height twice, once going up and once coming down.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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LESSON 1 (40 minutes (1 lesson)): Forming Quadratic Expressions from Areas: Mama Wanjiru's 40 m Fence

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners meet the anchoring phenomenon of Mama Wanjiru's 40 m fence, predict how area changes with width, and learn to write a width $\times$ length area as an algebraic expression and expand it to reveal an x^2 term, recognising it as a quadratic expression.
Knowledge	By the end of the lesson the learner should be able to form quadratic expressions: (a) write the area of a rectangle as width $\times$ length using x for the width, (b) expand $x(20 - x) = 20x - x^2$ and $(x + 3)(x + 5) = x^2 + 8x + 15$, and (c) recognise an expression whose highest power is 2 as a quadratic expression. This is part of Sub-strand 1.3 Quadratic Expressions and Equations (KICD Core Mathematics Grade 10, July 2025). Verify the exact KICD outcome wording against the KICD page.
Skills	Predicting from a pattern, modelling a plot with string loops and grid paper, building area tables, using variables to generalise (x and $20 - x$), expanding brackets, and communicating reasoning in groups. NGSS-style practices embedded: asking questions, developing and using models, analysing and interpreting data, using mathematics to represent a phenomenon.
Attitudes	Curiosity about why a fixed fence gives equal areas for different widths; willingness to test predictions and revise them; respect for each member's ideas in group work; responsibility in handling shared materials. Values and PCIs to be quoted verbatim from the KICD page, which has not been verified here.
Key Inquiry Question	1. How do we form quadratic expressions and equations? 2. Where do we apply quadratic equations in real-life situations? (KICD key inquiry questions; verify wording against the KICD page.) Lesson question: How does the area of Mama Wanjiru's plot change as the width changes, and why does an x^2 term appear?
Purpose in Storyline	Lesson 1 launches the phenomenon. Mama Wanjiru in Kericho has exactly 40 m of wire for a rectangular sukuma wiki and spinach plot. Her children Chebet and Kiprop notice that widths of 8 m and 12 m both give 96 m ² , 10 m gives 100 m ² , and widths above 20 m are impossible. Learners build the expression $x(20 - x)$, see the x^2 term appear, and open the DQB with the question 'Why does the area follow a pattern with x^2 ?'. Later lessons will use the expressions formed here (the lesson map names the area models and the x^2 term). Nothing is solved here beyond seeing the pattern.
Safety Notes	Cutting string and paper: use blunt-ended scissors and keep them on desks. Keep walkways clear while groups spread out grid paper. No outdoor work is needed. Make sure every learner can see and reach the materials and that seating allows all to participate.

B. LESSON OVERVIEW

Teacher introduces Mama Wanjiru's problem: 40 m of wire for a rectangular plot beside her house in Kericho. Learners first predict, individually and then in pairs, how the area changes as the width changes (Predict). In groups they model the plot with 40 cm string loops (1 cm = 1 m) on 1 cm grid paper, record widths, lengths and areas in a table, and compare with the class table (Observe). They discover that 8 m and 12 m give the same 96 m² because they are the same 8 m by 12 m rectangle turned round, that 10 m gives the maximum 100 m², and that widths above 20 m are impossible. They then write the length as $20 - x$, the area as $x(20 - x)$, expand it to $20x - x^2$, and use tiles to expand $(x + 3)(x + 5) = x^2 + 8x + 15$, naming both as quadratic expressions (Explain). The class opens the DQB (DQB Creation) and sketches Model v0: a diagram with a fixed perimeter, width x , length $20 - x$ and the question marks about equal areas and the maximum (Model Building). Nothing in this lesson solves for x ; solving is saved for later lessons.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Multiplying binomials: area model Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Represent rectangle measurements Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners listen to the story: 'Mama Wanjiru in Kericho has exactly 40 m of wire to fence a rectangular plot for sukuma wiki and spinach beside her house.' Individually they write a prediction in their books: 'If she makes the plot 2 m wide, 10 m wide, then 20 m wide, what will happen to the area?' They sketch a rough graph or sentence for how area changes as width increases. In pairs they compare predictions and mark with a star the width they think gives the biggest area. Some will predict that wider always means bigger; some will predict a middle width is best.	Draw a rectangle on the board and say: 'Mama Wanjiru has exactly 40 m of wire. Where does the wire go?' (Take 2 answers; confirm the wire goes all the way round, so length + width = 20 m.) Say: 'Take two minutes of silent thinking. Write your own prediction before talking.' WAIT TIME 10-15 seconds after asking: 'If the plot is 2 m wide, how long is it?' Cold-call 3 learners for the length. Then ask: 'Which width do you think gives the biggest area? Write it down and give a reason.' Cold-call 4 pairs without telling them if they are right. Record all predictions in a column on the board headed 'Our predictions'. Say plainly: 'We will not decide who is right by opinion. We will test.' Do not reveal the area table yet.	Predict-Observe-Explain (POE) with public prediction recording. Writing a prediction before seeing evidence commits learners to an idea, so that comparing it with the measured area table creates cognitive conflict and a need to explain. Recording all predictions on the board gives the class a shared reference to revisit in the Explain phase.	Observable indicators: (1) learners correctly state length = 20 – width, or at least that length and width add to 20 m (look at written predictions); (2) predictions include a reason, not just a number; (3) learners can say what changes and what stays fixed (perimeter fixed at 40 m). Note and tally learners who predict that the widest plot gives the largest area; plan to ask them about it in the Observe phase. Walk the room and tick a checklist during the pair talk.
Observe Phase  VIDEO: Two-step equation word problem: garden Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Compare area and perimeter Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	In groups of 4, learners get a 40 cm string loop (1 cm stands for 1 m), 1 cm grid paper and a table: width x length 20 – x area. They shape the loop into rectangles on the grid paper with widths 2, 4, 5, 6, 8, 10, 12, 14, 15, 16, 18 cm, read the length, count squares (or multiply) for the area and fill in the table. Each group is asked to try width 20 and widths above 20. They notice: 8 and 12 both give 96; 6 and 14 both give 84; 4 and 16 both give 64; 10 gives 100, the largest; at 20 the length is 0, so there is no plot; above 20 it cannot be made. They compare with their predictions. They are told that Chebet and Kiprop found the 8 m and 12 m equal areas on their own. A group shows that an 8 by	Hand out materials and say: 'One centimetre stands for one metre. Your loop is Mama Wanjiru's whole 40 m of wire. Every rectangle must use the whole loop.' Demonstrate one width ($x = 2$, length 18, area 36) and then let groups work. Circulate and ask: 'How did you get the length without measuring it?' and 'What do the width and length always add up to?' After 8 minutes ask groups to fill in a class table on the board, one row per group. WAIT TIME 10-15 seconds, then ask: 'What do you notice about the area column?' Cold-call 4 learners for noticing statements. Ask: 'Chebet and Kiprop say 8 m and 12 m both give 96 m ² . Are those two different plots?' Wait 10 seconds. Cold-call	Data table and pattern noticing (noticing and wondering). Learners generate their own data with a physical model, then look for patterns across rows (symmetry about 10, a maximum of 100, impossible widths). Producing the data themselves makes the later algebra a way of explaining something they have already seen, not an abstract rule.	Observable indicators: (1) learners fill in length as 20 – width without re-measuring after 2 or 3 rows; (2) area values in tables match the anchor table (36, 64, 75, 84, 96, 100, 96, 84, 75, 64, 36); (3) learners describe the symmetry in their own words (pairs of widths adding to 20 share an area); (4) learners explain that width 20 gives no plot. Check group tables for errors and ask the group to recount squares, not just correct them yourself. Record which groups notice the maximum at 10.

	12 plot and a 12 by 8 plot are the same rectangle turned round.	2 learners, then state plainly: 'It is the same 8 m by 12 m rectangle turned round, not two different shapes.' Ask: 'What happens at width 20, and what about 22?' Cold-call 3. Note that a 4 m by 24 m plot also has area 96 m ² but needs 56 m of wire, so it does not fit this fence. Return to the predictions on the board and ask learners to mark which were supported.		
Explain Phase  VIDEO: Comparing areas and perimeters of rectangles Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Represent rectangle measurements Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners ask: 'Can we write a rule for any width, not just the ones we tried?' With teacher guidance they let width = x m, length = (20 - x) m, and area A = x(20 - x). They expand using the grid: x(20 - x) = 20x - x ² , seeing that one term has x squared, since x times x is a square of side x. They check it with x = 8: 20 × 8 - 64 = 96. They then use area tiles (an x ² tile, x strips, unit squares) to model a different rectangle (x + 3) by (x + 5) and expand it: x ² + 5x + 3x + 15 = x ² + 8x + 15. They learn that an expression whose highest power is 2 is a quadratic expression, and they classify examples: 20x - x ² , x ² + 8x + 15, 3x + 7 (not quadratic), x ² (quadratic), x ³ + 2 (not quadratic).	Say: 'Our table only covers some widths. If Chebet says 9.5 m, how would we find the area quickly?' WAIT TIME 10-15 seconds, then cold-call 3 learners. Write on the board: width = x, length = ?, area = ? and ask learners to complete it in pairs (1 minute). Cold-call 2 pairs for the length 20 - x and the area x(20 - x). Ask: 'Use what you did in Grade 9 to expand x(20 - x). Which term is new?' WAIT 10 seconds. Cold-call 3 learners; write 20x - x ² and circle the x ² term. Ask learners to test x = 8 and x = 12 in 20x - x ² and say what they notice (both give 96). Next, draw a large rectangle with sides (x + 3) and (x + 5) divided into four parts and ask groups to label the area of each part (x ² , 5x, 3x, 15). Ask: 'What do we get when we add the parts that are alike?' Cold-call 3 for x ² + 8x + 15. Introduce the term 'quadratic expression' and give the rule: the highest power of x is 2. Give 5 quick examples for thumbs up or down. Say plainly that we have only written the area as an expression; we have not yet solved for any width.	Concrete-representational-abstract progression with an area model. The string loop and table (concrete) lead to the grid and tile diagrams (representational) and then to the algebraic expression (abstract). The same rectangle is shown in all three forms, so learners see that the x ² term comes from the square-shaped region of the diagram and is not an arbitrary algebra rule.	Observable indicators: (1) learners write length as 20 - x and area as x(20 - x) without help; (2) learners expand x(20 - x) to 20x - x ² and not 20 - x ² or 20x + x ² ; (3) learners label all four regions of the (x + 3) (x + 5) diagram correctly and combine 5x + 3x = 8x; (4) learners correctly sort quadratic from non-quadratic expressions (a thumbs up or down check on 5 examples, aim for at least 80% correct). Follow up privately with learners who give 20 - x ² and ask them to re-draw the grid. Use a 2-question exit slip: expand x(12 - x) and (x + 2)(x + 4).
Driving Question Board (DQB)	Learners write questions that this lesson has raised on sticky notes or slips, one question per slip, and	Put up the DQB chart paper with the driving question in the centre. Say: 'Think about what you still	Driving Question Board with question clustering. Collecting learners' own questions gives	Observable indicators: (1) each learner contributes at least one question; (2) at least some

Creation  VIDEO: Multiplying binomials: area model Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Compare area and perimeter Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	stick them on the class DQB under the heading: 'How can we find the plot dimensions when fence length and area are both fixed, and why do two different widths sometimes give the same area?' Possible learner questions: Why does the area go up and then come down? Why does 10 m give the biggest area? Why do 8 and 12 give the same area? Can we find the width for a given area without trial and error? Why does the x^2 appear? What if Mama Wanjiru had 60 m of wire? They sort questions into groups such as 'Why equal areas?', 'Why a maximum?', 'How to find widths for a target area?', and 'What is x^2?'	wonder. Write one question per slip. You have 3 minutes.' WAIT TIME 10-15 seconds before circulating. Give each learner at least one slip and cold-call 5 learners to read their question aloud before sticking it up. Ask: 'Which of these questions might we be able to answer with the x^2 idea?' and cold-call 2 learners. Help the class cluster the questions and label each cluster with a heading. Highlight the question the storyline gives for this lesson: 'Why does the area follow a pattern with x^2?' If no learner has asked it, add it yourself and say 'I wondered this too.' Keep the DQB visible and say that later lessons will return to it. Do not answer questions now.	them ownership of the unit, makes unresolved puzzles visible, and gives the class a way to track what has been explained later. Clustering shows learners how separate questions relate to the same underlying idea.	questions refer to the phenomenon (equal areas, the maximum, the fixed fence) and not only to procedures; (3) learners can say which cluster their question belongs to and why. Note learners who write only 'how do I expand?' type questions and prompt them to link to the plot. Photograph or copy the DQB so the class can compare it with the final DQB.
Model Building Phase  VIDEO: Two-step equation word problem: garden Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Area & perimeter of rectangles word problems <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners draw an initial class model, Model v0, in their books and on a large sheet: a rectangle labelled with width x and length $20 - x$, the fixed 40 m perimeter, and the area expression $x(20 - x) = 20x - x^2$. Beside it they add a small table of widths and areas (2 to 18 m) and arrows marking what they know (equal areas at 8 and 12, the maximum at 10, no plot at 20) and question marks for what they cannot yet explain (why equal, why a maximum, how to find widths for a target area). In pairs they write a one-sentence 'What I know so far' and 'What I still need to find out'.	Say: 'Our model must show what stays the same and what changes when the width changes.' Ask: 'What stays fixed? What changes?' WAIT TIME 10-15 seconds, then cold-call 3 learners. Build Model v0 on the board using learner suggestions and make sure it has: the rectangle, x, $20 - x$, the 40 m perimeter, the expression $20x - x^2$, and the table. Ask learners to put question marks where the model cannot yet explain the observations. Cold-call 3 pairs for their 'still need to find out' sentence and link each to a DQB cluster. Close with: 'Today we wrote area as an expression with an x^2. Next time we will use area models to see why squares behave the way they do.' Collect the exit slip (expand $x(12 - x)$ and $(x + 2)(x + 4)$) as learners leave. Keep Model v0 on the wall for revision in later lessons.	Cumulative model building with explicit gaps. Learners draw and label a model that captures only what the evidence so far supports, and mark what it cannot explain. Showing gaps in the model gives purpose to later lessons and teaches learners that models are revised as evidence grows.	Observable indicators: (1) the model shows a rectangle with x and $20 - x$ labelled and the perimeter 40 m stated; (2) the area expression is written correctly as $x(20 - x) = 20x - x^2$; (3) at least one question mark is placed where the model cannot yet explain a pattern (for example, the equal areas or the maximum); (4) exit slip: at least 80% of learners expand $x(12 - x) = 12x - x^2$ and $(x + 2)(x + 4) = x^2 + 6x + 8$ correctly. Use exit slip results to decide who needs extra practice with expanding before Lesson 2.

D. TEACHER REFLECTION

After the lesson ask yourself: (1) Did learners fill in the length as $20 - x$ with understanding, or just copy the pattern? (2) Could learners say plainly that 8 m by 12 m and 12 m by 8 m are the same plot? Who still thinks they are different shapes? (3) Did anyone confuse $x(20 - x) = 20x - x^2$ with $20 - x^2$? What did the grid diagram do to help? (4) Which learners predicted that the widest plot has the largest area, and did the table change their minds? (5) Did the DQB contain questions that link to the phenomenon, or mostly procedure questions? (6) Did groups run short of grid paper, string, or time, and what will you change? (7) Which learners need extra support with expanding brackets before Lesson 2? Note two changes for next time.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What we observed: With 40 m of wire, length + width = 20 m. Widths of 8 m and 12 m both give 96 m ² (the same 8 m by 12 m plot turned round), a width of 10 m gives the largest area of 100 m ² , and widths of 20 m or more do not give a plot.
What did I learn?	What we learned: If the width is x m then the length is $(20 - x)$ m and the area is $x(20 - x) = 20x - x^2$. Expanding $(x + 3)(x + 5)$ gives $x^2 + 8x + 15$. An expression whose highest power is 2 is a quadratic expression.
How does this explain the phenomenon?	What we can explain so far: The x^2 term appears because multiplying a width x by a length that also contains x makes an x by x square region in the area model, so the area is a quadratic expression. We cannot yet explain why areas repeat in pairs, why the maximum is at 10 m, or how to find widths for a target area without trial and error; these questions go on the DQB.

LESSON 2 (40 minutes (one lesson; extend to a double period if learners cut and paste their own tile sets)): Deriving the Squares Identities from Area: Why $12^2 = (10 + 2)^2$ and $8^2 = (10 - 2)^2$

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use area tiles and cut-out squares to derive the identities $(a + b)^2 = a^2 + 2ab + b^2$ and $(a - b)^2 = a^2 - 2ab + b^2$, then test them on the squares $12^2 = (10 + 2)^2 = 144$, $8^2 = (10 - 2)^2 = 64$ and $10^2 = 100$, which link to the plot widths in Mama Wanjiru's fencing problem.
Knowledge	By the end of the lesson the learner should be able to derive the quadratic identities $(a + b)^2 = a^2 + 2ab + b^2$ and $(a - b)^2 = a^2 - 2ab + b^2$ from the concept of area. An identity is true for all values of a and b. Learners also see how the identities fit the area models and expansions from Lesson 1.
Skills	Constructing and using area models (SEP: Developing and Using Models); making and testing a prediction with numbers (SEP: Analyzing and Interpreting Data); communicating a general pattern in words and symbols (SEP: Constructing Explanations); cutting and measuring accurately.
Attitudes	Core competencies: Critical Thinking and Problem Solving (learners explain why a middle term $2ab$ appears); Communication and Collaboration (learners justify their models to peers). Values: Integrity (record the cut and measured areas honestly, even when a check does not match at first); Responsibility (care for shared tiles, scissors and grid paper). PCI: Education for Sustainable Development, using scrap paper, old calendars and cardboard from the immediate environment for tiles.
Key Inquiry Question	1. How do we form quadratic expressions and equations? 2. Where do we apply quadratic equations in real-life situations? (KICD wording to be verified against the KICD page.) Lesson focus: How can area pictures help us write a squared bracket as a quadratic expression, and what do they tell us about Mama Wanjiru's plots?
Purpose in Storyline	Lesson 1 showed that the area of Mama Wanjiru's plot, $x(20 - x) = 20x - x^2$, is a quadratic expression, and that 8 m and 12 m both give 96 m ² . This lesson gives learners a tool for squared brackets, $(a + b)^2$ and $(a - b)^2$, that they will need to see why a square is the biggest possible plot. It starts Model v1, area models for expansions. Lesson 3 adds $a^2 - b^2 = (a - b)(a + b)$ and Lesson 4 runs expansion backwards to factorise. This lesson does not factorise and does not solve equations.
Safety Notes	Scissors and rulers: cut away from the body and carry them point-down. Use paper, card or old calendars only, with no blades or craft knives. Keep the floor clear of offcuts, and collect scraps in a bag for reuse or recycling. Learners with limited hand control may use pre-cut tiles or a partner to cut.

B. LESSON OVERVIEW

Mama Wanjiru in Kericho has 40 m of wire. Her children Chebet and Kiprop found that widths of 8 m and 12 m both give 96 m², while a 10 m by 10 m square gives 100 m². Learners already know the plot area $x(20 - x) = 20x - x^2$ is quadratic. In this lesson they ask: what happens when we square a bracket such as $(10 + 2)^2$ or $(10 - 2)^2$? Learners first predict 12^2 and 8^2 from the square of 10. Then they build a $10 + 2$ square from tiles (a 10×10 square, two 10×2 strips and a 2×2 square) and a $10 - 2$ square by cutting away from a 10×10 square. They write the general identities, test them on 12^2 , 8^2 and 10^2 , and record the pictures as Model v1. 12 m and 8 m are the length and width of the same rectangle, the 8 m by 12 m plot turned round. The squares are used only as number tests of the identities. A key conclusion is that $(a + b)^2$ is not $a^2 + b^2$: the middle strips $2ab$ are what the shortcut misses. Learners add questions to the Driving Question Board (DQB) and start Model v1.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Squaring binomials of the form (x+a)² Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Multiply perfect squares of binomials Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners recall Lesson 1: Mama Wanjiru's plot has area $x(20 - x) = 20x - x^2$, and widths of 8 m and 12 m both give 96 m². Each learner writes individually, on a sticky note or in their exercise book, a prediction for: (1) 12^2 if they only know $10^2 = 100$; (2) 8^2 if they only know $10^2 = 100$; (3) is $(10 + 2)^2$ equal to $10^2 + 2^2$? Yes or no, and why? They record a prediction and a reason in 'I think ... because ...' form. Then they pair-share and mark each prediction with a tick or a question mark.	Display on the board: 'Mama Wanjiru's plot can be 10 m wide, or 12 m, or 8 m. We know $10^2 = 100$.' Say: "Without a calculator, write down what you think 12^2 is, and tell me why." Give WAIT TIME of 10-15 seconds of silent thinking before any talking. Say: "Now turn to your partner and compare. Do not tell me who is right yet." Cold-call 3 pairs. Record every prediction, right or wrong, on the board without comment. Ask: "Who thinks $(10 + 2)^2 = 10^2 + 2^2 = 104$? Raise a hand. Who disagrees?" Count hands and write the numbers on the board. Say: "We will test this with real areas today." Do not give the answer.	Prediction with Reasoning (Predict-Observe-Explain): Writing a prediction first exposes the common misconception that $(a + b)^2 = a^2 + b^2$. Learners hold a clear claim to test, so the tile evidence in the Observe phase is a real test, not a lecture.	Observe and note on a class tally: (1) how many learners predict $12^2 = 100 + 4 = 104$ (the $a^2 + b^2$ misconception); (2) whether reasons mention area, squares or 'adding'; (3) learners who leave the reason blank or say 'just guessing'. Those who predict 104 are flagged for the middle strips. Indicator: at least 80% of learners write a prediction with some reason before the Observe phase begins.
Observe Phase  VIDEO: Multiplying binomials: area model Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Multiply perfect squares of binomials Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	In groups of 4, learners get a pack of paper tiles (cut from old calendars or exercise-book covers): one 10 cm by 10 cm square (a^2), two 10 cm by 2 cm strips (ab), and one 2 cm by 2 cm square (b^2). 1 cm stands for 1 m in the plot model. Part A, $(a + b)^2$: Learners fit the four pieces together to make a bigger square of side 12 cm, then measure the side, record that $10 + 2 = 12$, and count or calculate the total area as $100 + 20 + 20 + 4 = 144$ cm². They compare with $12 \times 12 = 144$ and with their prediction. Part B, $(a - b)^2$: Each group takes a fresh 10 cm by 10 cm square on grid paper and marks a border strip 2 cm wide along the top and along the right edge. They cut away the L-shaped border (two 10×2 strips overlapping at a 2×2 corner), leaving an 8 cm by 8 cm square.	Distribute tile packs and say: "Handle the scissors with care and cut away from your body." Model the first placement under the document camera or on the board: "Start with the big square, then add the two strips, then the little corner square. What shape do you get?" Circulate to every group within the first 5 minutes. Ask: "Where is the 2×2 piece in the picture? Why is it needed?" Then ask: "How many strips of 10×2 do you have? Why two and not one?" WAIT TIME 10-15 seconds before taking answers. For Part B, say: "When you cut away both strips, what did you take away twice?" Cold-call 2 groups to show the overlapping corner. Remind groups to record honestly: "If your measurement is 143 or 145, write what you measured and tell me why you think it is slightly off."	Area Model Construction (physical modelling): Making a bigger square from four labelled pieces, then a smaller square by cutting from a larger one, lets learners see the middle terms $2ab$ as real strips. Learners build meaning from the geometry before the algebra is written.	Check that each group: (1) names each tile piece as a^2 (10×10), ab (10×2), ab (10×2) and b^2 (2×2); (2) arrives at 144 cm² from pieces and from 12×12; (3) arrives at 64 cm² for the 8 cm square after cutting; (4) points to the overlapped corner in Part B. Indicator: at least 3 of 4 learners per group can explain where the two ab strips are in the picture. Note groups that count 104 and stop; guide them with 'Did the strips fit?'

	They count the area left: 64 cm^2. They compare with $100 - 20 - 20 + 4 = 64$ (the corner 2×2 was removed twice, so it is added back once). They fill in a table: square, side, area from pieces, area from side $\times$ side. Also, they fill in a final line: side 10, area 100.	Collect results from all groups onto a class table on the board. Link back: "12 m and 8 m are the length and width of the same plot turned round, and 10 m wide is the square."		
Explain Phase  VIDEO: Squaring binomials of the form $(x+a)^2$ Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Factorization using Perfect Square Trinomials (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners move from numbers to letters. In their groups they replace 10 with a and 2 with b, and write the area of the big square in two ways: as $(a + b)^2$ and as the sum of the pieces $a^2 + ab + ab + b^2 = a^2 + 2ab + b^2$. They write the same for the cut-out square: $(a - b)^2 = a^2 - 2ab + b^2$. They then test the two identities on 12^2, 8^2 and 10^2: $(10 + 2)^2 = 100 + 40 + 4 = 144$, $(10 - 2)^2 = 100 - 40 + 4 = 64$, and $10^2 = 100$ (b = 0, the middle strips vanish). They write a short answer to: 'Why is $(10 + 2)^2$ not $10^2 + 2^2$?' Finally they connect to the phenomenon in a one-line statement: 'The 12 m width and 8 m width are the same plot turned round; their squares are tested here only as numbers.' They also try with x: $(x + 3)^2 = x^2 + 6x + 9$, and compare it with Lesson 1's $(x + 3)(x + 5) = x^2 + 8x + 15$.	Ask: "If the big side is a and the small side is b, what are the areas of the four pieces?" Write answers as learners give them. WAIT TIME 10-15 seconds, then cold-call 4 learners for the four areas. Say: "An identity is true for ALL values of a and b, not only 10 and 2. How could we check that?" Invite 2 learners to try different numbers, such as a = 5 and b = 1: $(5 + 1)^2 = 36$ and $25 + 10 + 1 = 36$. Ask: "In the 12 m case, where is the extra 40 in $144 = 100 + 40 + 4$?" Expect: the two strips. Then say: "So who was right about 104?" Return to the Predict tallies and revisit them. For the minus identity ask: "Why does the b^2 come back with a plus sign?" Expect: the corner was cut away twice, so it is added back once. Briefly link: "In Lesson 1 we expanded $(x + 3)(x + 5)$. Today's identity is the special case where both brackets are the same." Do NOT introduce $a^2 - b^2$ (Lesson 3) or factorisation (Lesson 4).	Claim-Evidence-Reasoning with generalisation: Learners state the general claim (the identity), give evidence (the tile counts and the tests with 144, 64 and 100) and give reasoning (the middle strips, the double-cut corner). Moving from a specific number to letters builds the abstraction step by step.	Collect each group's identity statements and the answer to 'why not $10^2 + 2^2$?'. Look for: (1) both identities written correctly; (2) $2ab$ explained by the two strips; (3) $+ b^2$ in the minus identity explained by the double-cut corner; (4) all three tests (144, 64, 100) calculated correctly. Indicator: at least 75% of learners write a correct explanation. Quick check on whiteboards: "Expand $(x + 3)^2$." Expect $x^2 + 6x + 9$. Common error: $x^2 + 9$ or $x^2 - 6x + 9$. Reteach with a tile picture for these learners.
Driving Question Board (DQB) Creation  VIDEO: Multiplying binomials: area model Source: Khan Academy (English - US curriculum)  Search ARES	Learners add sticky notes to the class DQB under the driving question about dimensions of the plot, fence length and area, and equal areas. They place new notes on: 'What we figured out today' (e.g. 'Squaring a bracket gives a middle term $2ab$, not just $a^2 + b^2$.) and 'What we still wonder' (e.g. 'Can we cut a small square from a big square and rearrange it?', 'How	Say: "Take one sticky note. Write one thing we learned about squares and one question that is still open. You have 2 minutes." Circulate and read over shoulders. Cold-call 4 learners to read a note aloud and place it on the board. Ask: "Is this a question we can answer from area models, or do we need a new idea?" WAIT TIME 10-15 seconds before taking	Driving Question Board (questioning and tracking): Writing, sorting and ticking questions shows learners their growing understanding and keeps the storyline coherent. The open questions point toward the next lesson's investigation.	Scan the DQB notes: (1) at least 80% of learners add a note; (2) questions refer to the phenomenon, equal areas or the maximum, not only to procedures; (3) learners tick questions they can now answer. Indicator: at least 5 questions that relate to the next lessons (difference of squares, factorising, the 8 and 12 m solutions). If questions are only 'Is

for similar videos EXERCISE: Factorization using Perfect Square Trinomials Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	do we go backwards, from $x^2 + 8x + 15$ to brackets?', 'Why is the biggest area exactly when the plot is a square?'). Each learner writes at least one question. Learners group related questions and mark the ones this lesson answered, such as 'Why do squares have a middle term?', with a tick.	responses. Point out good questions that link to the phenomenon: "Several of you asked why the square gives the biggest area. We will come back to that, and in Lesson 8 we will see why 100 m^2 is the maximum." Move answered questions to a 'Partly answered' column. Do not answer the future-lesson questions now; just say which lesson will take them up (difference of squares is next lesson; going backwards is Lesson 4).		it on the exam?', prompt: "What would you like to know about Mama Wanjiru's plot?"
Model Building Phase VIDEO: Factoring Perfect Square Trinomials: A Sample Application <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar videos EXERCISE: Polynomial special products: perfect square <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners start Model v1 on a large chart: area models for expansions. They draw (1) the Lesson 1 area rectangle for $x(20 - x) = 20x - x^2$ with a short label, (2) the $(a + b)$ square split into a^2, ab, ab, b^2, with $a = 10$ and $b = 2$ and area 144, (3) the $(a - b)$ square with the cut border, $a = 10$ and $b = 2$ and area 64, and (4) the square with side 10 and area 100. Under each picture they write the identity or expression. They add a line 'Plots 8 m and 12 m are the same rectangle turned round; 10 m by 10 m is the square' and a 'Not yet explained' box with two questions from the DQB. Each learner copies Model v1 into their own exercise book. Then each learner writes one sentence in a Model Prompt: 'Our model shows ... It does not yet explain ...'.	Say: "Our model is a first draft. It will change in the next lessons. What must be on it so far?" WAIT TIME 10-15 seconds. Cold-call 3 groups to suggest one element each. Model the layout under the document camera: pictures at the top, identities in the middle, 'Not yet explained' at the bottom. Say: "Make sure each picture has its numbers labelled: 10, 2, 144, 64, 100." Circulate and check labels. Ask: "Where in your model can I see why 12^2 is bigger than 10^2?" (The extra strips and corner are added.) "Where can I see why 8^2 is smaller?" (The border strips are removed, with the corner returned once.) Collect the 'does not yet explain' statements and read 2 aloud. Close with a preview: "Next lesson we cut a small square out of a large one and rearrange the pieces, and the $52^2 - 48^2$ puzzle will follow." Do not write $a^2 - b^2$ on the model yet.	Cumulative Model Revision (Model v1): Building the first version of a class model that is revised in later lessons shows learners how scientific and mathematical ideas grow. Naming what the model cannot yet explain keeps the driving question open and motivates the next lesson.	Check each model for: (1) the three area pictures drawn and labelled correctly; (2) both identities written correctly; (3) the statement that 8 m and 12 m are the same plot turned round; (4) at least one honest 'not yet explained' item. Indicator: at least 80% of learners can complete the model prompt sentence correctly. Exit ticket on one half-page: "Expand $(x + 4)^2$ using a tile picture or the identity." Expect $x^2 + 8x + 16$. Sort the tickets into correct, middle term missing and sign error, and plan a short reteach at the start of Lesson 3.

D. TEACHER REFLECTION

After the lesson, consider: (1) How many learners moved from ' $a^2 + b^2$ ' to the full $a^2 + 2ab + b^2$? What in the tile evidence changed their minds? (2) Did groups correctly explain why b^2 is added back in $(a - b)^2$? If not, would a coloured overlap help next time? (3) Did all learners link the identities back to Mama Wanjiru's plot and

understand that 8 m and 12 m are the same rectangle turned round? Did anyone say they were 'different shapes'? (4) Did every learner get a turn to handle tiles, or did a few dominate? How will you re-balance roles in Lesson 3? (5) Which DQB questions need to be addressed in the next lesson, and which could confuse learners if not handled with care? Record 2 actions for Lesson 3.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We built a 12 cm square from a 10×10 square, two 10×2 strips and a 2×2 square, and found its area is 144. We cut a 2 cm border from a 10 cm square and left an 8 cm square of area 64. The 10 cm square has area 100.
What did I learn?	An identity is true for all values. $(a + b)^2 = a^2 + 2ab + b^2$, and $(a - b)^2 = a^2 - 2ab + b^2$. The middle term $2ab$ comes from two strips, so $(a + b)^2$ is not $a^2 + b^2$. In $(a - b)^2$, the b^2 is added back because the corner was cut away twice.
How does this explain the phenomenon?	The squares $12^2 = (10 + 2)^2 = 144$, $8^2 = (10 - 2)^2 = 64$ and $10^2 = 100$ agree with the identities. The 8 m and 12 m widths are the same plot turned round. Model v1 shows area pictures for expansions, and it does not yet explain why 100 m^2 is the largest area or how to find both widths for a target area.

LESSON 3 (40 minutes (one lesson; Predict 5, Observe 12, Explain 10, DQB 5, Model 8)): Cutting a Small Square from a Large Square: Difference of Two Squares

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners derive the identity $a^2 - b^2 = (a - b)(a + b)$ by cutting a small square from a large square and rearranging the pieces into a rectangle. They then use it for quick mental calculation ($52^2 - 48^2 = 4 \times 100 = 400$) and see how it links back to the Kericho plot, where the 40 m fence gives length $20 - x$ and area $x(20 - x)$. The class cumulative model is revised so that it holds all three identities.
Knowledge	By the end of the lesson the learner should be able to: (1) derive $a^2 - b^2 = (a - b)(a + b)$ from an area model; (2) state, with the other two identities, $(a + b)^2 = a^2 + 2ab + b^2$ and $(a - b)^2 = a^2 - 2ab + b^2$; (3) use $a^2 - b^2 = (a - b)(a + b)$ to calculate quickly, for example $52^2 - 48^2 = (52 - 48)(52 + 48) = 4 \times 100 = 400$; (4) see, as a preview only, that $100 - (x - 10)^2 = (10 - (x - 10))(10 + (x - 10)) = (20 - x)(x) = x(20 - x)$, the area of the Kericho plot.
Skills	Developing and using models (NGSS SEP 2): cutting and rearranging paper squares and drawing area diagrams. Analysing and interpreting data (SEP 4): comparing the direct calculation $2704 - 2304$ with the identity method. Constructing explanations (SEP 6): explaining why the L-shaped region becomes a rectangle with sides $(a - b)$ and $(a + b)$. Using mathematics (SEP 5): substituting numbers into an identity. KICD core competencies: Critical thinking and problem solving; Communication and collaboration; Creativity and imagination.
Attitudes	Learners show Responsibility by handling scissors carefully and keeping paper offcuts for the cumulative model. They show Integrity by cutting and measuring accurately. They show Respect by listening to each partner's explanation. PCI: Education for Sustainable Development, by reusing old exercise-book paper, and Life Skills, by making quick mental estimates in daily trade at a Kericho kiosk.
Key Inquiry Question	How do we form quadratic expressions and equations? Where do we apply quadratic equations in real-life situations? (KICD key inquiry questions, to be verified against the KICD page.) Lesson question: How can cutting a small square out of a large square give us a shortcut for calculating, and how does it connect to the area of Mama Wanjiru's plot?
Purpose in Storyline	Lesson 1 taught how to write a width $\times$ length area as an algebraic expression and expand it to see an x^2 term. Lesson 2 derived $(a + b)^2$ and $(a - b)^2$ from area tiles and cut-out squares. Lesson 3 adds the third identity, $a^2 - b^2 = (a - b)(a + b)$, by cutting one square out of another. The class model of the plot is revised to include all three identities. The link $100 - (x - 10)^2 = x(20 - x)$ is only previewed here, since its full use in explaining the maximum of 100 m^2 comes in Lesson 8. Lesson 4 will run expansion backwards to factorise $x^2 + bx + c$.
Safety Notes	Scissors: cut away from the body, pass them handle first, and keep them on the desk when not in use. Cut only along ruled lines on paper. Paper offcuts should be collected in a tray so no one slips on them. Use no string loops this lesson.

B. LESSON OVERVIEW

Anchoring phenomenon: Mama Wanjiru in Kericho has exactly 40 m of fencing wire for a rectangular vegetable plot (sukuma wiki and spinach), so length + width = 20 m. Her children Chebet and Kiprop found that widths of 8 m and 12 m both give 96 m^2 , that 10 m gives the largest area of 100 m^2 , and that widths above 20 m are impossible. So far the class has written the area as $x(20 - x) = 20x - x^2$ (Lesson 1) and built $(a + b)^2$ and $(a - b)^2$ from area models of squares, testing $12^2 = 144$, $8^2 = 64$ and $10^2 = 100$ (Lesson 2). In this lesson learners predict what remains when a small square is cut from a corner of a large square, then cut, rearrange and measure the pieces to discover that the L-shape becomes a rectangle with sides $(a - b)$ and $(a + b)$. They apply the result to $52^2 - 48^2 = 4 \times 100 = 400$ as a mental shortcut, and see a preview that $100 - (x - 10)^2 = x(20 - x)$: the plot area is a 10 m square with a small square taken away. The class revises its cumulative model so it holds all three identities.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Factoring difference of squares: two variables (example 2) Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Difference of squares intro Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners recall the plot model from Lessons 1 and 2. Each learner silently writes a prediction in their exercise book: 'A square of side 10 cm has a square of side 4 cm cut from one corner. What is the area left? Can you think of a way to write it as one product of two lengths?' They calculate $100 - 16 = 84 \text{ cm}^2$ and sketch where they think the L-shape could be cut into a rectangle. Then, as a mental-maths challenge, they predict how long it would take to work out $52^2 - 48^2$ without a calculator and give a guess of the answer. Many will square both numbers ($2704 - 2304$); this is recorded as their starting strategy.	Draw the 10 cm square with a 4 cm corner removed on the board. Say: 'Remember Mama Wanjiru's plot in Kericho. In Lesson 2 we used squares of 12, 8 and 10 to test our tiles. Today we ask what happens when a small square is taken from a big one.' Give 2 minutes of silent writing. Then say: 'Write your prediction, then draw how you would cut this L-shape.' WAIT TIME: 10 seconds after each question before anyone speaks. Cold-call 3 learners to share different cut ideas without judging them. Then write on the board: $52^2 - 48^2 = ?$ Ask: 'Without a calculator, how would you do it?' Cold-call 2 learners and record their methods. Say: 'Keep your predictions, we will test them with scissors.'	Predict-Observe-Explain (POE) with Elicit Prior Ideas. Writing a prediction first commits learners to an idea that evidence can later confirm or change. Recording the 'square both numbers' strategy creates a need for a shortcut, so the identity is discovered as a solution to a real problem.	Observable indicators: (1) does the learner calculate $100 - 16 = 84$ correctly? (2) does the sketch show any attempt to rearrange the L-shape? (3) does the learner recall from L2 that squares can be modelled with areas? (4) how many learners use the 'square both numbers' method for $52^2 - 48^2$? Tally on the board. Learners who cannot find the L-shape area get a grid-paper square to count unit squares.
Observe Phase  VIDEO: Factoring difference of squares: analyzing factorization Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Difference of Two Squares (PLIX) Source: CK-12  Search ARES for similar readings	In pairs, learners use old exercise-book paper and rulers. Step 1: draw and cut out a large square of side $a = 10 \text{ cm}$. Step 2: from one corner mark and cut out a small square of side $b = 4 \text{ cm}$. Label the remaining L-shape with $a^2 - b^2$ and calculate its area, $100 - 16 = 84 \text{ cm}^2$. Step 3: cut the L-shape along a straight line (a dotted line is drawn from the inner corner to the outside edge) into two rectangles: one measuring 10 cm by 6 cm and one measuring 4 cm by 6 cm. Step 4: rotate the 4 cm by 6 cm rectangle and join it to the 10 cm by 6 cm rectangle so that the pieces form one long rectangle. Measure its sides: 6 cm ($a - b$) by	Distribute scrap paper, rulers and scissors and review safety: 'Cut away from your body, pass scissors handle first.' Circulate and ask each pair: 'What is the area of the piece you cut away? What is left?' Prompt: 'Which sides of your new rectangle are made from a and b ? How do you know?' Do not give the formula. WAIT TIME: 10 seconds after each prompt. If a pair is stuck, ask: 'Where did the strip of width 6 cm come from on the big square?' After the first rectangle is found, ask groups to try different a and b values. Cold-call 3 pairs to read their table rows aloud and write them on the board so everyone can see $84 = 6 \times 14$	Modelling with physical manipulatives and Pattern Finding across Cases. Learners cut and rearrange the shape (SEP 2) and compare a table of examples (SEP 4) to move from specific numbers to a general statement. Moving between the concrete model, the table and the symbols builds a connected understanding that the identity comes from area and is not just a rule to memorise.	Observable indicators: (1) are the pieces cut accurately (check measured sides of 6 cm and 14 cm)? (2) can the learner say which side of the rectangle is $(a - b)$ and which is $(a + b)$? (3) does the second example row in the table match the pattern? (4) can the learner write $a^2 - b^2 = (a - b)(a + b)$ in their own words? Use a quick checklist on a clipboard. Learners who struggle are given a pre-drawn grid square and asked to shade the L-shape.

	14 cm (a + b). Area = $6 \times 14 = 84$ cm², the same as the L-shape. Step 5: repeat with a second pair of squares of their own choosing (for example a = 9 and b = 5, giving $4 \times 14 = 56 = 81 - 25$) and record both in a table with columns a, b, $a^2 - b^2$, (a - b), (a + b), (a - b)(a + b). Finally learners write the general rule in symbols.	and $56 = 4 \times 14$. Ask: 'What do you notice about the two lengths of the rectangle and the numbers a and b?'		
Explain Phase  VIDEO: Factoring difference of squares: two variables - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Difference of squares Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners first explain the identity: in pairs, using the diagram, they write why the L-shape area $a^2 - b^2$ equals the rectangle area (a - b)(a + b), noting that the same paper was only rearranged so the area is unchanged. They then compare the three identities side by side: $(a + b)^2 = a^2 + 2ab + b^2$, $(a - b)^2 = a^2 - 2ab + b^2$ and $a^2 - b^2 = (a - b)(a + b)$. Next they use the identity for mental calculation: $52^2 - 48^2 = (52 - 48)(52 + 48) = 4 \times 100 = 400$, and check against $2704 - 2304 = 400$. They try $75^2 - 25^2 = 50 \times 100 = 5000$ and $105^2 - 95^2 = 10 \times 200 = 2000$ (Kericho tea-buying context: 105 kg and 95 kg of tea). Finally, as a preview only, they are shown a 10 by 10 square standing for the largest plot, with a small square of side (x - 10) removed: $100 - (x - 10)^2 = (10 - (x - 10))(10 + (x - 10)) = (20 - x)(x)$. They check with x = 8: $100 - 4 = 96 = 8 \times 12$, and with x = 6: $100 - 16 = 84 = 6 \times 14$. The full use of this link comes in Lesson 8.	Ask: 'In your own words, why does the rectangle have the same area as the L-shape?' WAIT TIME: 15 seconds. Cold-call 3 learners for explanations and invite 2 more to add or challenge. Then write the three identities on the board and ask: 'Which one did we find today and which picture matches each one?' Pose: 'Use the shortcut for $52^2 - 48^2$. Who can show it on the board?' Invite one learner to show the steps and ask the class to confirm with $2704 - 2304$. Give a minute for pairs to try $75^2 - 25^2$ and $105^2 - 95^2$. Cold-call 2 pairs. Common misconception to raise: 'Is $52^2 - 48^2$ the same as $(52 - 48)^2$?' Let learners compute 16 versus 400 and discuss. Preview carefully: 'This is just a peek. Take x = 8 on Mama Wanjiru's plot: $100 - (8 - 10)^2 = 100 - 4 = 96$. Do you see 8×12 hiding here? We will use it fully in Lesson 8.' Make clear the 8 m and 12 m are the same rectangle turned round.	Claim-Evidence-Reasoning (CER) with Contrasting Cases. Learners state the claim (the identity), supply evidence (cut paper and numerical checks), and give reasoning (area is conserved when pieces are rearranged). Contrasting $a^2 - b^2$ with $(a - b)^2$ exposes a common error and sharpens understanding. The preview with x = 8 and x = 6 connects the identity to the anchoring phenomenon without pre-empting Lesson 8.	Observable indicators: (1) does the explanation mention that the same pieces were rearranged so area stays the same? (2) can the learner compute $52^2 - 48^2 = 400$ using the identity and not by squaring? (3) does the learner avoid confusing $a^2 - b^2$ with $(a - b)^2$? (4) can the learner verify $100 - (x - 10)^2 = x(20 - x)$ for x = 8 giving 96? Use mini-whiteboards: each learner shows their answer for $75^2 - 25^2$ (5000) at the same time and the teacher scans the room.
Driving Question Board (DQB) Creation  VIDEO: Special products of the form (x+a)(x-a)	Learners add to the Driving Question Board. They write new questions on sticky notes, for example: 'Does $a^2 - b^2$ work for any numbers, even decimals?', 'Why is the plot area always 100 minus a square?', 'Can we use these identities to work out the	Point to the DQB and say: 'Today we found a shortcut. What new questions has it raised about Mama Wanjiru's plot?' Give 2 minutes of silent writing and then 2 minutes to post notes. WAIT TIME: 10 seconds before calling on anyone. Cold-call 3 learners to	Driving Question Board with Question Sorting. Posting and sorting questions keeps learners as owners of the inquiry and shows that each lesson narrows the gap between the evidence so far and the original puzzle: how to find widths for a target area	Observable indicators: (1) do notes connect to the plot or the identities rather than being random? (2) does at least one question move towards solving for x? (3) can learners say which questions are partly answered? Count the notes per category and

Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Difference of squares intro Source: Khan Academy (English - US curriculum) Search ARES for similar readings	widths for a target area?', 'Does $a^2 + b^2$ also factorise?' They also move earlier questions that are now answered, such as 'Why does 8 m give the same area as 12 m?' (partly: the identity suggests symmetry about 10, but the full answer comes in Lesson 6), to a 'Partly answered' column. Each pair places at least one note on the board.	read their notes aloud and sort them into groups: 'about identities', 'about the plot', 'about solving'. Say: 'Which of these will we answer in the next lessons?' Mark the note about 'a target area without trial and error' with a star and say we will begin working towards it in Lesson 4 with factorising.	without trial and error.	record which pairs posted only recall questions so the teacher can prompt them next lesson.
Model Building Phase VIDEO: Factoring difference of squares: two variables (example 2) Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Difference of Two Squares (PLIX) Source: CK-12 Search ARES for similar readings	Learners revise the class cumulative model on chart paper. The model started in Lesson 2 with area models of squares (version 1). Learners now add a third panel showing the large square with a small square cut from one corner and the rearranged rectangle labelled $(a - b)$ by $(a + b)$. Under the three panels they write the three identities in the exact forms: $(a + b)^2 = a^2 + 2ab + b^2$, $(a - b)^2 = a^2 - 2ab + b^2$ and $a^2 - b^2 = (a - b)(a + b)$. They add a small link box showing the plot facts: fence 40 m, length = $20 - x$, area $x(20 - x)$, 8 m and 12 m both 96 m², 10 m the largest at 100 m², with the note '$100 - (x - 10)^2 = x(20 - x)$? Check in Lesson 8'. Each learner then completes a 3-line summary in their own book.	Say: 'Our model has to explain Mama Wanjiru's plot. What did we have before today, and what do we add now?' Invite 3 learners to draw the three panels, with cold-calls for the labels. WAIT TIME: 10 seconds after asking 'Which identity belongs under which picture?' Ask: 'What can our model not yet do?' Lead learners to say that it cannot yet find the widths for a target area, so we need to factorise and solve (Lessons 4 to 6). Close with an exit ticket on a half-sheet: 'Use $a^2 - b^2 = (a - b)(a + b)$ to find $63^2 - 37^2$, and write one sentence about how it links to a square and the Kericho plot.' (Answer: $26 \times 100 = 2600$.) Collect the tickets.	Cumulative Model Revision with Gaps Identified. Adding the third identity to the model shows learners how the evidence is growing from lesson to lesson. Naming what the model still cannot do gives the next lesson a purpose and keeps every phase tied to the anchoring phenomenon.	Observable indicators on the exit ticket: (1) correct answer 2600 using $(63 - 37)(63 + 37) = 26 \times 100$; (2) the three identities written correctly in the model; (3) a sentence linking to area of squares or to the plot. Sort tickets into 3 piles (secure, partly secure, not yet). The 'not yet' pile gets a short re-cut of the paper squares at the start of Lesson 4.

D. TEACHER REFLECTION

1) Did learners discover the rearrangement themselves, or did they need the cut line suggested? Note which pairs needed support. 2) How many learners still treated $a^2 - b^2$ as $(a - b)^2$? Plan a short mini-lesson with numbers (16 versus 400) if more than a quarter did. 3) Was the preview of $100 - (x - 10)^2 = x(20 - x)$ kept light, so that the full explanation was left for Lesson 8? 4) Did all learners say plainly that widths 8 m and 12 m give the same 8 m by 12 m rectangle turned round? 5) Which mental-calculation example ($52^2 - 48^2$, $75^2 - 25^2$, $105^2 - 95^2$) most engaged learners, and could a local trading context be added next time? 6) Were scissors handled safely and was waste paper collected?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We cut a small square from the corner of a large square and rearranged the L-shape into a rectangle. The rectangle had sides $(a - b)$ and $(a + b)$ and the same area as the L-shape, for example 10 cm and 4 cm gave $6 \times 14 = 84 \text{ cm}^2$. We also saw that $52^2 - 48^2 = 4 \times 100 = 400$ matches $2704 - 2304 = 400$, and that for the Kericho plot, $x = 8$ gives $100 - 4 = 96 = 8 \times 12$.
What did I learn?	The identity $a^2 - b^2 = (a - b)(a + b)$ is true for all values of a and b . It gives a quick way to calculate differences of squares mentally. Together with $(a + b)^2 = a^2 + 2ab + b^2$ and $(a - b)^2 = a^2 - 2ab + b^2$, it makes up the three identities in our model. The plot area $x(20 - x)$ can be written as $100 - (x - 10)^2$, a preview we will use fully in Lesson 8.
How does this explain the phenomenon?	Cutting the L-shape and rearranging the pieces does not change the area, so $a^2 - b^2$ must equal $(a - b)(a + b)$. This helps explain Mama Wanjiru's plot in Kericho: the area is 100 minus a small square, which hints at why widths on either side of 10 m give equal areas (8 m and 12 m are the same 8 m by 12 m plot turned round) and why 10 m gives the most, 100 m^2 . We still need to learn how to factorise and solve to find widths for a target area without trial and error.

LESSON 4 (40 minutes (one lesson; Predict 6 min, Observe 12 min, Explain 10 min, DQB 5 min, Model 7 min)): Factorising $x^2 + bx + c$: Running the Expansion Backwards to Find the Plot

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners run the expansion $(x + p)(x + q) = x^2 + (p + q)x + pq$ in reverse. They factorise $x^2 + bx + c$ by finding two numbers whose sum is b and whose product is c , and check each answer with area tiles or by expanding. They then link $x^2 - 20x + 96 = (x - 8)(x - 12)$ back to Mama Wanjiru's 8 m and 12 m plots.
Knowledge	By the end of the lesson the learner should be able to: (1) explain that factorising means writing an expression as a product; (2) state that in $x^2 + bx + c = (x + p)(x + q)$, $p + q = b$ and $p \times q = c$; (3) factorise expressions such as $x^2 + 8x + 15$ and $x^2 - 20x + 96$, including cases where b and c are negative; (4) verify a factorisation by arranging tiles or by expanding; (5) link $x^2 - 20x + 96 = (x - 8)(x - 12)$ to the plot widths 8 m and 12 m, which are the same 8 m by 12 m rectangle turned round. (KICD Specific Learning Outcome: factorise quadratic expressions.)
Skills	Analysing and interpreting data: look for the pattern between the numbers in an expansion and the coefficients. Using mathematics and computational thinking: list factor pairs of c and test their sums. Developing and using models: build a tile rectangle to represent $x^2 + bx + c$. Constructing explanations: justify why the sum and product rule works using the area model. Core competencies: Critical thinking and problem solving; Communication and collaboration; Learning to learn.
Attitudes	Values: Responsibility (taking care of shared tile sets and handing them back complete) and Respect (listening to each group member's strategy before correcting). Learners also develop persistence when a factor pair does not work, and appreciation that algebra can unlock real Kenyan farming problems without trial and error. PCIs: Life skills (decision making when choosing between candidate number pairs); Education for Sustainable Development (using locally made cardboard tiles).
Key Inquiry Question	How do we form quadratic expressions and equations? Where do we apply quadratic equations in real-life situations? Lesson focus: If we know the area expression $x^2 + bx + c$ of a rectangle, how can we find its width and length without trial and error?
Purpose in Storyline	Lessons 1 and 2 showed that the plot area is a quadratic expression and that expanding brackets gives an x^2 term. Lesson 4 turns the expansion around, which is a key tool for the central puzzle. Here learners discover that $x^2 - 20x + 96 = (x - 8)(x - 12)$, because -8 and -12 have sum -20 and product 96 . The numbers 8 and 12 are exactly the two widths that gave 96 m^2 in the plot table. Lesson 5 extends factorising to $ax^2 + bx + c$, and Lesson 6 uses the zero-product property to solve $x^2 - 20x + 96 = 0$ properly. This lesson does NOT yet solve the equation formally; it only builds the factorising skill and plants the clue.
Safety Notes	Cardboard tiles have sharp corners, so cut them with blunt-nosed scissors and round off the corners. Keep tiles and string off the floor so no one trips. Collect all tile sets at the end of the lesson and count them back in.

B. LESSON OVERVIEW

Mama Wanjiru's children, Chebet and Kiprop, noticed that widths of 8 m and 12 m both enclose 96 m^2 with her 40 m of wire in Kericho. Today learners find out why those two numbers appear. They predict the dimensions of a rectangle from an area expression, build it with x^2 tiles, x strips and unit squares, and discover the sum-and-product pattern. They factorise $x^2 + 8x + 15$ (from Lesson 1) and then $x^2 - 20x + 96$, whose factors $(x - 8)$ and $(x - 12)$ contain the two plot widths. Learners check every answer by expanding. The lesson ends with Driving Question Board (DQB) updates and a revision of the cumulative model, which already holds the area models (Lesson 2) and all three identities (Lesson 3), so that it now records the factorised form of the plot-area equation.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Factoring Quadratic Expressions with Negative Coefficients: A Sample Application Source: CK-12  Search ARES for similar videos  EXERCISE: Quadratics by factoring (intro) Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners sit in groups of 4. The teacher writes on the board: 'Mama Wanjiru's new bed has area $x^2 + 8x + 15$ m ² . Its width is $(x + 3)$ m. What is its length?' Individually, then in groups, learners write a prediction on a sticky note or in the exercise book: the length, and one sentence explaining how they got it. Some will guess by division, others by recalling the Lesson 1 expansion $(x + 3)(x + 5) = x^2 + 8x + 15$. Next, learners predict a harder case: 'If the area is $x^2 - 20x + 96$, what two brackets multiply to give it? What do you notice about 96 and the plots from the story?' Groups record their prediction and their confidence level (sure, unsure, guessing) for each question. Link to the phenomenon: the 96 m ² is the area that both the 8 m and 12 m widths gave.	Say: 'Remember Mama Wanjiru's plot in Kericho. In Lesson 1 we expanded $(x + 3)(x + 5)$. Today we run it backwards. Before I teach anything, I want your best prediction.' Write the first expression and give 1 minute of silent individual thinking. Then WAIT TIME: 10-15 seconds after asking 'What would the missing length be?' before taking any response. Cold-call 3 learners (not volunteers) to share predictions and ask each 'What made you think that?' Do not say whether answers are right. Write the second expression $x^2 - 20x + 96$ and ask: 'Where have we seen the number 96 before?' WAIT 10-15 seconds, then cold-call 2 learners. Record all predictions on a chart headed 'Our predictions'. Circulate and note which groups use division, trial and error or recall of the expansion.	Predict-Observe-Explain (POE) with a confidence rating. Committing to a prediction in writing, together with a confidence level, creates a cognitive stake. When learners later see the tile evidence they notice conflicts between their prediction and the result. Linking 96 to the plot table activates prior knowledge from Lesson 1 and shows that the 8 m and 12 m plots are connected to the expression.	Look for: (1) how many groups write a length of $(x + 5)$ and whether they give a reason; (2) whether learners link the 96 to the 8 m and 12 m plots; (3) whether any learner mentions numbers that add to the middle term and multiply to the last term. Tally on a class checklist: Ready (gives a reason using expansion), Developing (guesses without reason), Needs support (cannot start). Use the tally to form mixed-ability tile groups.
Observe Phase  VIDEO: Strategy in factoring quadratics (part 1 of 2) Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Quadratics by factoring (intro) Source: Khan Academy (English - US curriculum)  Search ARES	Groups receive a tile set: 1 x^2 tile (a square of side x), 10 x strips (x by 1) and 20 unit squares, made from cardboard, with x about 5 cm and 1 unit = 1 cm. Task A ($x^2 + 8x + 15$): build a rectangle using exactly 1 x^2 tile, 8 x strips and 15 unit squares. Learners discover that the x strips split into 3 and 5 and the units form a 3 by 5 block, giving sides $(x + 3)$ and $(x + 5)$. They record the numbers 3 and 5 and fill a table: sum = 8, product = 15. Task B ($x^2 + 7x + 12$ and $x^2 + 9x + 20$): learners repeat the build and look for the rule. They then list factor pairs of c , test which pair adds to b , and record p , q , $p + q$	Distribute tile sets with the instruction: 'Take care of these tiles; they belong to the whole class. Count them before and after.' Say: 'Build $x^2 + 8x + 15$ as a single rectangle. No gaps, no overlaps. Then tell me the two side lengths.' Circulate and ask each group: 'Where are your 8 strips? How did you split them?' WAIT TIME 10-15 seconds before any hint. Cold-call 2 groups to show their tile rectangle to the class. Give the table prompt: 'What do you notice about the numbers on the sides, 3 and 5, and the numbers in the expression, 8 and 15?' WAIT 10-15 seconds, then	Concrete-Representational-Abstract (CRA) with Pattern Finding. Learners first use physical tiles (concrete), then complete a table of p , q , sum and product (representational), then state the rule in symbols (abstract). Because they run Lesson 1's expansion in reverse, they see that the rule $p + q = b$ and $p \times q = c$ is the same pattern read backwards. Pattern finding in a table lets learners generate the rule themselves rather than being told it.	Observable indicators: (1) learner builds a closed rectangle with no gaps and names both sides; (2) the table has correct p , q , sum and product values for at least two expressions; (3) learner lists factor pairs of 96 systematically and identifies 8 and 12; (4) learner expands $(x - 8)(x - 12)$ correctly to verify. Use a quick check: ask each group to show a mini-whiteboard with the factorisation of $x^2 + 9x + 20$. Note groups that show $(x + 4)(x + 5)$ as ready and those with wrong signs or swapped values as needing a prompt.

for similar readings	and $p \times q$ in a table. Task C (negative numbers): the teacher shows a pre-drawn chart of $x^2 - 20x + 96$ with the 'minus' tiles drawn in red. Learners list the factor pairs of 96 (1×96, 2×48, 3×32, 4×24, 6×16, 8×12) and test which pair adds to 20. They discover $8 + 12 = 20$ and $8 \times 12 = 96$, so with both negative: $-8 + -12 = -20$ and $(-8)(-12) = 96$. Therefore $x^2 - 20x + 96 = (x - 8)(x - 12)$. Learners then expand $(x - 8)(x - 12)$ to confirm: $x^2 - 12x - 8x + 96 = x^2 - 20x + 96$. Finally, learners notice that 8 and 12 are the same widths that gave 96 m^2 in the plot table.	cold-call 3 learners. For Task C say: 'Mama Wanjiru's children found 8 m and 12 m both give 96. List all the factor pairs of 96. Which pair adds to 20?' Support learners who confuse signs by asking: 'If the product is positive and the sum is negative, what must be true about both numbers?' Ask groups to expand their answer as a check before moving on. Watch for learners who find pairs with the right product but ignore the sum.		
Explain Phase  VIDEO: Strategy in factoring quadratics (part 2 of 2) Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Use Square Roots to Solve Quadratic Equations (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners work in their groups to write a short 'Factorising Recipe' using their evidence: Step 1: look at c and list pairs of numbers that multiply to c. Step 2: choose the pair that adds to b. Step 3: write $(x + p)(x + q)$. Step 4: check by expanding. They then apply the recipe to new expressions on a worksheet: $x^2 + 10x + 21$, $x^2 + 6x + 8$, $x^2 - 9x + 20$, $x^2 - 7x + 10$ and $x^2 + 2x - 15$. For $x^2 + 2x - 15$, learners must work out that one number is negative: 5 and -3 give sum 2 and product -15, so $(x + 5)(x - 3)$. Each learner explains to a partner why the signs behave as they do. Finally, each group answers in writing: 'How does factorising $x^2 - 20x + 96$ as $(x - 8)(x - 12)$ help explain why the 8 m and 12 m plots have the same area?' Expected idea: the numbers 8 and 12 add up to 20, which is the fence half-length, so a width of 8 m gives a length of 12 m and a width of 12 m gives a length of 8 m. These are the same 8 m by 12 m rectangle turned round, not two	Say: 'Use your tile evidence to write the recipe in your own words. Which step would you teach a Grade 9 learner first?' WAIT TIME 10-15 seconds after posing 'Why must both numbers be negative when the sum is negative and the product is positive?' Cold-call 3 learners to answer, then ask a fourth to restate. For $x^2 + 2x - 15$ ask: 'What is different about the product here? What does that tell you about the signs of p and q?' Cold-call 2 pairs to share. State plainly: 'The plots with widths 8 m and 12 m are the SAME rectangle turned round. Their widths add up to 20 m, half of the 40 m of wire.' Do not formally solve $x^2 - 20x + 96 = 0$ or introduce the zero-product property; say: 'In Lesson 6 we will use these brackets to solve the equation. Today we only factorise.' Address the misconception that $(x + 3)(x + 5) = x^2 + 15$ by asking the learner to expand it with tiles. Collect 3 worksheets to check accuracy before the lesson ends.	Self-Explanation and Peer Teaching (Turn-and-Talk). Writing a recipe in their own words forces learners to organise the procedure and the reason behind it. Explaining signs to a partner exposes gaps in reasoning. Connecting the factorisation back to 8 m and 12 m moves the lesson from a procedure to an explanation of the phenomenon: the numbers in the brackets are the plot widths that give 96 m^2.	Observable indicators: (1) the recipe includes both the sum and product conditions and a check step; (2) learner correctly factorises at least 4 of 5 expressions, including at least one with a negative c; (3) in their written answer learners state that $8 + 12 = 20$ and $8 \times 12 = 96$, and that the 8 m and 12 m plots are the same rectangle turned round; (4) no learner claims the two plots are different shapes. Collect exit slips with the factorisation of $x^2 - 11x + 24$ (answer $(x - 3)(x - 8)$). Sort slips into Secure, Nearly and Not yet for the next lesson's grouping.

	different shapes. Learners should recognise that the factors give the two widths but are not asked to solve the equation formally yet.			
Driving Question Board (DQB) Creation  VIDEO: Factoring Quadratic Expressions with Negative Coefficients: A Sample Application Source: CK-12  Search ARES for similar videos  EXERCISE: Use Square Roots to Solve Quadratic Equations Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners return to the class Driving Question Board with the main question: 'How can we use quadratic expressions and equations to find the dimensions of a plot when its fence length and area are both fixed, and why do two different plots sometimes have the same area?' Each group writes on sticky notes (a) one thing they can now answer, for example 'How do we find the two numbers in the brackets?' and (b) one new question, for example 'Can we use the brackets to find the width directly?', 'What if the number in front of x^2 is not 1?' or 'Does every expression factorise?'. They place answered notes under 'Answered so far' with the evidence (the tile picture or the sum-and-product table) and new questions under 'Still wondering'. The class sorts the notes into themes such as 'Solving the equation' (Lesson 6) and 'Harder expressions' (Lesson 5).	Say: 'Look at our DQB. Which question can we now answer with the evidence from our tiles?' WAIT TIME 10-15 seconds, then cold-call 3 groups to place one note each and justify it with evidence. Ask: 'What new question has factorising raised about Mama Wanjiru's plot?' Cold-call 2 learners. Point out the key unanswered question: 'We found brackets with 8 and 12 in them. How do we know that $x = 8$ and $x = 12$ are the widths? That is the job of Lesson 6.' Group related notes with a marker and label themes. Do not answer new questions now; promise when they will be addressed (Lesson 5: $ax^2 + bx + c$; Lesson 6: solving the equation).	Question Clustering on the Driving Question Board. Sorting questions into 'answered' and 'still wondering' makes progress visible and gives learners ownership of the learning path. Each answered note must cite evidence, so learners practise linking claims to data. New questions drive the sequence into Lessons 5 and 6.	Observable indicators: (1) at least one answered note per group cites tile or table evidence; (2) new questions are productive (about solving, about a not equal to 1, or about the maximum) rather than off-topic; (3) learners can say which lesson will likely answer their question. Record the number of groups that ask a question about turning the brackets into widths, as a sign that they are ready for Lesson 6.
Model Building Phase  VIDEO: Strategy in factoring quadratics (part 1 of 2) Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Quadratics by	Learners open their cumulative model of Mama Wanjiru's plot, which contains the v1 area models (Lesson 2) and all three identities (Lesson 3). In groups, they add a new section called 'Factorised form'. They copy the new relationship for the plot: Area = $x(20 - x) = 20x - x^2$; target 96 m^2: $x^2 - 20x + 96 = (x - 8)(x - 12)$, because $-8 + -12 = -20$ and $(-8)(-12) = 96$. They draw a simple diagram showing the 8 m by 12 m rectangle in two orientations, labelled 'width 8 m, length 12 m'	Say: 'Our model has area models and three identities. What new idea from today must go in?' WAIT TIME 10-15 seconds, then cold-call 3 learners to suggest additions. Check that the diagram shows 8 m by 12 m as ONE rectangle turned round, and that the widths add to 20 m. Ask: 'Why do the numbers 8 and 12 add to 20 in our model?' Cold-call 2 learners. Challenge: 'Our model shows factors of $x^2 - 20x + 96$. Does it yet tell us the solutions? What is still missing?' Accept 'zero-	Cumulative Model Revision. Learners revise a shared model rather than start a new one, which shows how each lesson adds a tool for explaining the phenomenon. Stating what the model can and cannot yet explain builds scientific habits of mind: models are improved by evidence and have limits. It also sets up Lesson 6, where the factorised form will be used to solve the equation.	Observable indicators: (1) the revised model contains $x^2 - 20x + 96 = (x - 8)(x - 12)$ with the correct sum -20 and product 96; (2) the diagram shows ONE 8 m by 12 m rectangle in two orientations; (3) learner names a limit (cannot yet solve for x or explain the maximum); (4) exit task: $x^2 + 12x + 35 = (x + 5)(x + 7)$ and $x^2 - 13x + 36 = (x - 4)(x - 9)$, with a correct expansion check for at least one. Learners who get both correct are Secure; those with sign errors get a short tile re-teach

taking square roots: strategy <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	and 'width 12 m, length 8 m', with a note: 'the same rectangle turned round; width + length = 20 m'. They also add the sum-and-product rule in a box: $x^2 + bx + c = (x + p)(x + q)$ where $p + q = b$ and $p \times q = c$, with the check 'expand to confirm'. Finally, each learner writes one sentence on what the model can now explain and one thing it cannot yet do (it cannot yet show that $x = 8$ and $x = 12$ are the solutions; it does not yet explain the maximum area of 100 m ²).	product property' only as a hint at Lesson 6; do not teach it. Give each learner a 2-question exit task: factorise $x^2 + 12x + 35$ and $x^2 - 13x + 36$, then check one by expanding. Collect the tasks. Remind learners to return all tiles.	at the start of Lesson 5.
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D. TEACHER REFLECTION

After the lesson, consider: (1) Did learners discover the sum-and-product rule from the tile patterns, or did I tell it too early? (2) Which learners listed factor pairs systematically, and which guessed? (3) Did any learners say the 8 m and 12 m plots are two different shapes, and how did I correct this? (4) How well did learners handle negative numbers, especially in $x^2 + 2x - 15$? (5) Did the link between $(x - 8)(x - 12)$ and the plot widths feel meaningful, or did it seem like a coincidence? (6) Did I resist solving the equation formally, leaving it for Lesson 6? (7) Did tile shortages or sharing problems affect the activity, and could locally made cardboard tiles help? (8) Which exit-slip errors will shape the opening of Lesson 5?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We built rectangles from x^2 tiles, x strips and unit squares. $x^2 + 8x + 15$ made a rectangle with sides $(x + 3)$ and $(x + 5)$. For $x^2 - 20x + 96$ we listed the factor pairs of 96 and found that $8 \times 12 = 96$ and $8 + 12 = 20$.
What did I learn?	To factorise $x^2 + bx + c$, find two numbers p and q whose sum is b and whose product is c , then write $(x + p)(x + q)$. Check by expanding. When the sum is negative and the product is positive, both numbers are negative.
How does this explain the phenomenon?	$x^2 - 20x + 96 = (x - 8)(x - 12)$ because $-8 + -12 = -20$ and $(-8)(-12) = 96$. The numbers 8 and 12 are the widths that gave 96 m ² on Mama Wanjiru's 40 m fence in Kericho. They add up to 20 m, so the 8 m and 12 m plots are the same 8 m by 12 m rectangle turned round.

LESSON 5 (40 minutes (one lesson)): Factorising $ax^2 + bx + c$ ($a \neq 1$) and Special Cases: Which Method Works Best?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners factorise $ax^2 + bx + c$ where $a \neq 1$ so that they can later solve target-area equations for Mama Wanjiru's 40 m fenced plot when the area expression has a number in front of x^2 .
Knowledge	Learners factorise $ax^2 + bx + c$ ($a \neq 1$) using the grouping method (split the middle term using two numbers whose product is ac and whose sum is b) and the area-tile method. They recognise $x^2 + 10x + 25$ as a perfect square $(a + b)^2$ and $x^2 - 100$ as a difference of two squares $(a - b)(a + b)$, and they check every answer by expansion. KICD strand outcome: the learner should be able to factorise quadratic expressions.
Skills	Splitting the middle term, grouping and taking out common factors, arranging area tiles into a rectangle, checking by expansion, comparing methods and justifying a choice, and communicating reasoning. NGSS-style practices: Developing and Using Models (tile and area models), Analysing and Interpreting Data (comparing methods), and Constructing Explanations.
Attitudes	Precision and integrity in checking work, perseverance when the first split fails, and respect for classmates' different methods. KICD values: Responsibility and Unity. KICD core competencies: Critical Thinking and Problem Solving, and Communication and Collaboration. PCI: Life Skills (decision making).
Key Inquiry Question	How do we form quadratic expressions and equations? Where do we apply quadratic equations in real-life situations? (Lesson focus: How do we factorise when the x^2 term has a number in front, and which method should we choose?) (The KICD key inquiry wording is to be verified against the KICD page.)
Purpose in Storyline	Lesson 4 gave the sum-and-product method for $x^2 + bx + c$. Real plots with extra rules often give an x^2 term with a number in front, so this lesson extends factorising to $a \neq 1$. It also links the special cases from Lessons 2 and 3 and holds a DQB review of which methods work best. Lesson 6 will use this factorising to solve equations such as $x^2 - 20x + 96 = 0$, which resolves the central puzzle.
Safety Notes	Learners use only paper tiles, grid paper, scissors and string loops. Scissors are used with care and the cutting is done at desks. Keep bags and string off the floor to avoid tripping. No hazardous materials are used.

B. LESSON OVERVIEW

Mama Wanjiru's children, Chebet and Kiprop, know that her 40 m of wire gives the area $x(20 - x)$, and in Lesson 4 the class factorised $x^2 + bx + c$ using sum and product. Now the class asks what to do when there is a number in front of x^2 , such as $2x^2 + 7x + 3$. Learners predict how many rectangles of tiles can be built, then build $2x^2 + 7x + 3$ from area tiles (two x^2 tiles, seven x strips, three unit squares) and find the rectangle $(2x + 1)$ by $(x + 3)$. They then use the grouping method ($ac = 6$, split $6x$ and x : $2x^2 + 6x + x + 3 = 2x(x + 3) + 1(x + 3) = (2x + 1)(x + 3)$) and check by expansion. Learners test the special cases $x^2 + 10x + 25$ (a perfect square) and $x^2 - 100$ (a difference of two squares, from Lessons 2 and 3). Finally the class reviews the Driving Question Board (DQB) to decide which method works best and when, and revises the cumulative model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Intro to grouping Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Perfect squares Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners recall Mama Wanjiru's plot from Kericho: 40 m of wire, width x m, length $(20 - x)$ m. The teacher writes $2x^2 + 7x + 3$ on the board and says that Chebet found a plot-style rectangle with this area. Individually, then in pairs, learners predict on their sheets: (1) Can sum and product from Lesson 4 factorise this directly? (2) What might the two sides of the rectangle look like? They write a first guess such as $(2x + 3)(x + 1)$ and keep it to test later.	Ask: "Mama Wanjiru's area $x(20 - x)$ has an x^2 term. What if the x^2 term has a number in front, like $2x^2$?" Write $2x^2 + 7x + 3$ on the board. Say: "Write your prediction silently first." Give WAIT TIME of 10-15 seconds, then 2 minutes of silent writing. Pair-share for 1 minute. Cold-call 4 learners to give a guess for the two brackets, writing every guess on the board without saying which is right. Ask: "What would you do to check a guess?" and take the answer "expand it".	Predict-Observe-Explain (prediction phase): committing to a guess before instruction makes learners notice the gap in Lesson 4's method, so they have a reason to learn a new method, and wrong guesses become evidence to compare later.	Observable indicators: (a) at least 10 of 12 sampled learners write a prediction instead of leaving the sheet blank; (b) learners who propose expanding to check show they remember Lesson 1 expansion; (c) the teacher notes on a tally how many guesses are of the form $(2x + p)(x + q)$, $(x + p)(x + q)$ or no guess. Learners who write $(x + 3)(x + 1)$ show they have not noticed the 2 in front of x^2.
Observe Phase  VIDEO: Factoring quadratics by grouping Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Perfect squares intro Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Part A (area tiles, about 8 minutes): in groups of 4, learners use cut paper tiles (an x^2 tile, an x strip, a unit square) to build $2x^2 + 7x + 3$ using 2 x^2 tiles, 7 x strips and 3 unit squares. They must arrange all tiles into one rectangle with no gaps and read the sides. They find the rectangle with sides $(2x + 1)$ and $(x + 3)$. They sketch it on grid paper. Part B (grouping, about 8 minutes): the teacher guides them to find $ac = 2 \times 3 = 6$ and two numbers with product 6 and sum 7 (6 and 1). They split the middle term: $2x^2 + 6x + x + 3$, group as $(2x^2 + 6x) + (x + 3)$, take out common factors $2x(x + 3) + 1(x + 3)$ and get $(2x + 1)(x + 3)$. They check by expansion: $2x^2 + 6x + x + 3 = 2x^2 + 7x + 3$. They then compare with their tile picture: the $6x + x$ strips are the two x-strip groups on the sides of the rectangle. Part C (special cases, about 5 minutes): learners test $x^2 + 10x + 25$ by tiles and by sum and product (5 and 5): $(x + 5)^2$ and link it to $(a + b)^2 = a^2 + 2ab + b^2$ from Lesson 2. They test $x^2 - 100$ with a missing middle term: $(x - 10)(x + 10)$, linking to $a^2 - b^2 = (a - b)(a + b)$.	Circulate with a checklist. Say: "Use all the pieces. No gaps, no overlaps. Read the length and width of your rectangle." Ask groups: "Where is the x^2 part of your rectangle? Where are the 7 strips?" For grouping, write $2x^2 + 7x + 3$ on the board and ask: "What is $a \times c$? Which two numbers multiply to 6 and add to 7?" Give WAIT TIME of 10-15 seconds before cold-calling 3 learners for the pair of numbers. Ask: "Why must we split $7x$ into $6x$ and x, not $5x$ and $2x$?" ($5x + 2x$ also adds to $7x$ but the grouping will not give a common bracket.) Have learners try 5 and 2 and see that the brackets do not match. Insist everyone checks by expansion. For the special cases, ask: "Which of these has no middle term? What does Lesson 3 tell us?" Cold-call 3 learners to explain.	Modelling with multiple representations: the same factorisation is shown as a tile rectangle, a grouping calculation and an expansion check. Matching the pieces across all three connects the algebra to the area of Mama Wanjiru's plot, so factorising is seen as finding the sides of a rectangle from its area.	Observable indicators: (a) groups form a complete rectangle with sides $(2x + 1)$ and $(x + 3)$ within 8 minutes; (b) learners write $ac = 6$ correctly and name 6 and 1; (c) learners show the check by expansion on paper; (d) learners recognise $x^2 + 10x + 25$ as $(x + 5)^2$ and $x^2 - 100$ as $(x - 10)(x + 10)$. Watch for a common error: leaving out the second group's factor of 1 and writing $2x(x + 3) + (x + 3)$ wrongly as $2x(x + 3)$. Prompt: "What is left when you take out $(x + 3)$ from $x + 3$?"

	b) from Lesson 3.			
Explain Phase  VIDEO: Strategy in factoring quadratics (part 1 of 2) Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Difference of squares intro Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners work in pairs on three new expressions: (1) $3x^2 + 10x + 3$ ($ac = 9$, numbers 9 and 1, giving $(3x + 1)(x + 3)$); (2) $2x^2 + 5x + 2$ ($ac = 4$, numbers 4 and 1, giving $(2x + 1)(x + 2)$); (3) $x^2 + 6x + 9$ and $4x^2 - 25$ as special cases, giving $(x + 3)^2$ and $(2x - 5)(2x + 5)$. Each pair factorises one expression by the area-tile or grouping method and one by the other method, and checks every answer by expansion. Then each pair writes a 'Method Card' explaining in two sentences when each method is useful. Finally they connect back to the phenomenon: $x^2 - 20x + 96 = (x - 8)(x - 12)$ from Lesson 4 can be done by sum and product (-8 and -12 add to -20, multiply to 96), and the class discusses whether grouping gives the same answer (split $-20x$ as $-8x - 12x$).	Say: "Choose a different method for each of your two expressions. Explain to your partner why you chose it." Give WAIT TIME of 10-15 seconds after each question. Cold-call 5 pairs in total, one for each expression and one for the Method Card. Ask: "How did you know your answer was right?" and expect "we expanded". Ask: "Is $(x + 3)^2$ the same as $x^2 + 9$? Show me with your tiles." Use the error to reinforce Lesson 2. For the phenomenon link, ask: "Mama Wanjiru's 96 m^2 equation was $x^2 - 20x + 96$. Does grouping give the same brackets as sum and product?" Support learners who struggle by giving a half-completed grouping frame. Extension: ask fast groups to factorise $6x^2 + 11x + 3$ ($ac = 18$, numbers 9 and 2, giving $(2x + 3)(3x + 1)$).	Peer explanation and Claim-Evidence-Reasoning: each pair states a claim (the factorised form), supplies evidence (expansion or tile picture) and gives reasoning (why the chosen method fitted). Explaining why a method was chosen builds strategic understanding and not only procedure.	Observable indicators: (a) at least 8 of 12 sampled pairs correctly factorise at least two of the three expressions; (b) every answer carries a visible expansion check; (c) the Method Card gives a reason such as "no middle term means difference of two squares" or "$a = 1$ means sum and product"; (d) learners say that factorising $x^2 - 20x + 96$ gives the 8 and 12 of the plot widths. Learners who cannot split the middle term receive the prompt "Which two numbers multiply to ac and add to b?" and a short list of factor pairs.
Driving Question Board (DQB) Creation  VIDEO: Difference of squares intro Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Difference of squares Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	The class returns to the Driving Question Board. Learners review the questions already posted, for example "Why do 8 m and 12 m give the same area?" and "How do we find widths without trial and error?" They add sticky notes with new questions or partial answers from today, such as "Which method works best, and when?" and "Can every quadratic be factorised?" Groups sort the methods into a table on the board: sum and product ($a = 1$), grouping ($a \neq 1$), area tiles (any, but helpful for seeing why), and the special cases (perfect square, difference of two squares). They place a sticky note on questions that today's lesson helped to answer, and mark the question "How do we find the widths for a target area?"	Say: "Look at the board. Which questions did factorising help us with today? Which questions are still open?" Give WAIT TIME of 10-15 seconds. Cold-call 4 learners to move or add a note and explain why. Ask: "Which method would you choose for $x^2 + 14x + 49$? For $5x^2 - 20$? For $3x^2 + 8x + 4$? Why?" Take a quick show of hands for each choice and ask 2 learners to defend theirs. Record the class's agreed rules of thumb: first look for a common factor or special case, then use sum and product if $a = 1$, otherwise use grouping, and use tiles to see or check. Say: "Next lesson we will use these factors to actually find the widths."	Driving Question Board review: sorting questions into answered, partly answered and open shows learners their growing understanding and keeps the Kericho plot problem as the purpose for each new skill. Comparing methods builds the habit of choosing a strategy.	Observable indicators: (a) each group adds at least one new question or partial answer; (b) learners justify a method choice with a feature of the expression (for example, "no middle term"); (c) learners identify 'solve $(x - 8)(x - 12) = 0$' as the next step without prompting; (d) the teacher records which questions learners think are still open. If fewer than half can justify their choice, repeat the choice task with two more expressions at the start of Lesson 6.

	as 'closer, needs solving in Lesson 6'.			
Model Building Phase  VIDEO: Intro to grouping Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Perfect squares Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners revise the cumulative class model. In their notebooks they add a new section to the model: 'Factorising turns an area into its sides.' They draw the $2x^2 + 7x + 3$ tile rectangle with sides $(2x + 1)$ and $(x + 3)$, write the grouping steps beside it, and add a flow chart: 'Look for a special case or common factor → if $a = 1$ use sum and product → if $a \neq 1$ use grouping or tiles → check by expansion.' They add a link to the phenomenon: $x^2 - 20x + 96 = (x - 8)(x - 12)$, where 8 and 12 are the plot widths that give 96 m^2 with the 40 m fence, and the 8 m and 12 m plots are the same 8 m by 12 m rectangle turned round. They write one sentence on what is still missing: "We have not yet used the factors to find x."	Say: "Add today's idea to our model. How does factorising help us find the sides of Mama Wanjiru's plot?" Give WAIT TIME of 10-15 seconds, then cold-call 3 learners for their sentence. Check that learners state plainly that widths 8 m and 12 m describe the same rectangle turned round and not two different shapes. Ask: "If $(x - 8)(x - 12) = 0$, what might x be?" Take guesses but do not confirm, saying "We will prove it in Lesson 6." Collect the notebooks of 6 learners to check the flow chart and the expansion checks. Assign a short practice: factorise $2x^2 + 5x + 3$ and $x^2 + 12x + 36$, and check both by expansion.	Cumulative model revision: adding factorising to the earlier area models and identities shows how the pieces fit together, and the 'still missing' sentence keeps the next step (solving) connected to the central puzzle.	Observable indicators: (a) the model includes the tile rectangle, the grouping steps and the expansion check; (b) the flow chart puts 'check special cases first' and 'check by expansion' in sensible places; (c) learners write that factorising gives the sides of the plot from its area; (d) the practice answers $(2x + 3)(x + 1)$ and $(x + 6)^2$ are correct. A learner who writes $(x + 6)^2 = x^2 + 36$ needs to revisit Lesson 2 with tiles.

D. TEACHER REFLECTION

After the lesson, check: Which learners could split the middle term without the tiles, and which relied on the tiles? How many learners checked by expansion without being reminded? Did anyone confuse $(x + 5)^2$ with $x^2 + 25$? Was the 8 m and 12 m plot explained as one rectangle turned round and not two shapes? Which method did learners prefer and did their reasons match the expression? What would you change in the grouping instruction or the Method Card task? Note any learners who need extra practice on factor pairs before Lesson 6, where factorising is used to solve $x^2 - 20x + 96 = 0$.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We built $2x^2 + 7x + 3$ from area tiles (two x^2 tiles, seven x strips, three unit squares) and found a rectangle with sides $(2x + 1)$ and $(x + 3)$. We also saw that $x^2 + 10x + 25$ makes a square and that $x^2 - 100$ has no middle term.
What did I learn?	To factorise $ax^2 + bx + c$ with $a \neq 1$, find two numbers with product ac and sum b , split the middle term, group, take out common factors, and check by expansion. Perfect squares and differences of two squares are special cases to look for first.
How does this explain the phenomenon?	Factorising turns an area back into the sides of a rectangle, just as $x^2 - 20x + 96 = (x - 8)(x - 12)$ gives the widths 8 m and 12 m for Mama Wanjiru's 96 m^2 plot. Which method is best depends on the expression: a special case, sum and product when $a = 1$, or grouping or tiles when $a \neq 1$. We still need to use the factors to find x, which is the next lesson.

LESSON 6 (40 minutes (one lesson; the KICD lesson allocation for this sub-strand should be verified against the KICD page)): Two Widths, One Area: Solving Quadratic Equations by Factorisation

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use the zero-product property to solve $x^2 - 20x + 96 = 0$ after factorising it, and interpret both solutions in the plot context. This resolves the central puzzle of the phenomenon: why widths 8 m and 12 m both give 96 m ² .
Knowledge	Learners will know that: (1) a quadratic equation is a quadratic expression set equal to 0, written in standard form $ax^2 + bx + c = 0$; (2) the zero-product property says if $p \times q = 0$ then $p = 0$ or $q = 0$; (3) $x(20 - x) = 96$ rearranges to $x^2 - 20x + 96 = 0$, which factorises as $(x - 8)(x - 12) = 0$, so $x = 8$ or $x = 12$; (4) the two solutions describe the same 8 m by 12 m rectangle turned round, because width + length = 20 m; (5) solutions must be checked against the context $0 < x < 20$.
Skills	Learners will be able to: rearrange an equation to zero, factorise it using the sum-and-product method from Lessons 4 and 5, apply the zero-product property to find both roots, check each root by substituting into $x(20 - x)$, and explain in words what each root means for Mama Wanjiru's plot. NGSS-style practices embedded: Analysing and Interpreting Data (comparing the area table with the roots), Constructing Explanations (why two roots), Developing and Using Models (revising the cumulative model).
Attitudes	Learners show perseverance when a product is not 0, curiosity about why two answers appear, and respect for peers' reasoning in pair discussions. Values: Responsibility (checking answers carefully), Unity (sharing methods in pairs). Core competencies: Critical Thinking and Problem Solving, Communication and Collaboration. Kenyan PCI: Life Skills (decision making in planning land use). KICD wording for outcomes, competencies, values and PCIs for this sub-strand must be copied verbatim from the KICD page when finalising, as the supplied OCR text does not contain this sub-strand.
Key Inquiry Question	KICD key inquiry questions (verify wording against the KICD page): 1. How do we form quadratic expressions and equations? 2. Where do we apply quadratic equations in real-life situations? Lesson focus question: How can we solve $x(20 - x) = 96$ without trial and error, and why do two widths appear?
Purpose in Storyline	Lessons 1 to 5 built the area expression $x(20 - x) = 20x - x^2$, the area models, the identities and factorising skills. In Lesson 6 learners turn the target-area question into an equation and solve it. This resolves the central puzzle of the phenomenon (8 m and 12 m both give 96 m ²) and revises the cumulative model to include two solutions. Lesson 7 will extend this to other real-life situations and rejecting invalid solutions, and Lesson 8 to design tasks and the maximum area.
Safety Notes	Low-risk classroom lesson. Use blunt-tipped scissors for any paper cut-outs and keep string loops and tiles off the floor to prevent tripping. Ensure bags are stored away from walkways. Be sensitive that some learners may find it hard to publicly state wrong answers; use pair-sharing first.

B. LESSON OVERVIEW

Mama Wanjiru in Kericho has exactly 40 m of wire for a rectangular vegetable plot of sukuma wiki and spinach, so width x m means length $(20 - x)$ m and area $x(20 - x)$. Her children Chebet and Kiprop found that widths 8 m and 12 m both give 96 m². Until now learners found this by trial and error from the area table. In this lesson they predict how many widths could give 96 m², observe how the equation $x(20 - x) = 96$ becomes $x^2 - 20x + 96 = 0$, factorise it to $(x - 8)(x - 12) = 0$ and use the zero-product property to get $x = 8$ or $x = 12$. They explain that both answers are the same 8 m by 12 m plot turned round (width and length add up to 20 m), check both against the context, add questions to the Driving Question Board and revise the cumulative model to include two solutions.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Solving Real-World Problems Using Quadratic Equations: A Sample Application Source: CK-12  Search ARES for similar videos  EXERCISE: Zero product property Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners sit in pairs with the area table from Lesson 1 hidden except for the row $x = 10$, $A = 100$. On a half-sheet they individually write: (a) how many different widths could give exactly 96 m^2 with the 40 m fence, and why; (b) a guess for each width. They then write one sentence on whether there could be a way to find the widths without trying numbers. Pairs share and write their prediction on a sticky note labelled 'My prediction about Mama Wanjiru's plot'.	Display: 'Mama Wanjiru wants the plot to enclose exactly 96 m^2 using all 40 m of wire. Chebet says one width works. Kiprop says two widths work. Who do you think is right?' Say: 'Write your own answer first. Do not talk yet.' Give WAIT TIME of 15 seconds before pairs speak, then 2 minutes of silent writing. Then say: 'Turn to your partner and compare for one minute.' Cold-call 4 pairs: 'Chebet's way or Kiprop's way, and what is your reason?' Record all predictions on the board in two columns (one width / two widths) without saying who is correct. Say: 'Hold on to your prediction, today we will test it with mathematics, not guessing.'	Predict-Share-Revisit: committing to a prediction in writing creates a cognitive stake, so learners pay attention to the evidence and notice when their thinking changes. Learners' reasons (e.g. 'the table goes up then down') reveal prior ideas about symmetry that can be built on in the Explain phase.	Observe: do learners reason from the area table pattern (rising then falling) or from guesses? Listen for the idea that 'the areas go up and come down' as evidence of symmetry. Count how many pairs predict two widths and note how many give a reason. Success indicator: at least 3 of the 4 cold-called pairs give a reason that refers to the table or to the sum width + length = 20 m.
Observe Phase  VIDEO: Zero product property Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Zero product property Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Step 1: Learners write the area equation: $x(20 - x) = 96$. In pairs they expand the bracket (Lesson 1) and rearrange to standard form, getting $20x - x^2 = 96$, so $x^2 - 20x + 96 = 0$. Step 2: Learners recall factorising (Lessons 4 and 5) and find two numbers whose sum is -20 and whose product is 96 (-8 and -12), writing $(x - 8)(x - 12) = 0$. They check by expanding. Step 3: With a teacher demonstration using strips, learners observe the zero-product property: they are given 'Mystery product cards' where $p \times q = 0$ (e.g. $7 \times ? = 0$, $? \times ? = 0$ with one factor 0) and list what must be true. They then apply it: $x - 8 = 0$ or $x - 12 = 0$, so $x = 8$ or $x = 12$. Step 4: Learners substitute both values into $x(20 - x)$: $8 \times 12 = 96$ and $12 \times 8 = 96$. They test a non-root, $x = 9$, in $(x -$	Hand out the observation table and mystery product cards. Say: 'We have a target area of 96 m^2 . Last time we guessed from a table. Today we will use an equation.' Ask: 'What is the length when the width is x ?' and cold-call 2 learners $(20 - x)$. Write $x(20 - x) = 96$ and ask: 'What must I do to get everything on one side equal to zero?' WAIT TIME 10 seconds. Cold-call 3 learners for each step of rearranging. For factorising say: 'Use the sum-and-product method from Lesson 4: which two numbers multiply to give 96 and add to give -20 ?' WAIT TIME 15 seconds, then pairs list factor pairs of 96 on mini-whiteboards. Check by expanding $(x - 8)(x - 12)$ with 2 volunteers. For zero-product, hold up the card ' $__ \times __ = 0$ ' and ask: 'If two numbers multiply to give 0,	Observation table with Claim-Evidence-Reasoning: each row records what was done (claim), the working (evidence) and why it is allowed (reasoning, e.g. 'zero-product property'). Contrasting product = 0 with product = 96 helps learners see why we rearrange to zero before splitting into factors, building the key idea instead of a memorised rule.	Check mini-whiteboards: 100% of pairs should write $x^2 - 20x + 96 = 0$ correctly. Look for the common error of writing $(x + 8)(x + 12)$ (sign error) or stopping at $(x - 8)(x - 12) = 96$. Success indicators: learners correct sign errors after the expansion check, at least 80% explain 'one of the brackets must be zero', and learners state that they rearranged to zero first. Record the names of learners who need a Lesson 4 re-teach card.

	$8)(x - 12) = 1 \times (-3) = -3 \neq 0$, to see that the product is zero only at the roots. They record all in an observation table: step, working, what it shows.	what can you say about at least one of them?' WAIT TIME 10 seconds, then cold-call 3 learners. Ask: 'Why can't we say the same for a product equal to 96? For example, if $p \times q = 96$, does p have to be 8?' Walk around and ask 'How do you know it works?' Prompt substitution: 'Put $x = 8$ and then $x = 12$ into $x(20 - x)$. What do you get?' Also test $x = 9$: 'Is $(x - 8)(x - 12)$ zero?'		
Explain Phase  VIDEO: Solving Real-World Problems Using Quadratic Equations: A Sample Application Source: CK-12  Search ARES for similar videos  EXERCISE: Quadratics by taking square roots: strategy <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners work in groups of 3 to produce a short written explanation answering three prompts: (1) Why does the equation give two answers? (2) What do $x = 8$ and $x = 12$ mean for the plot? (3) Are they two different shapes? They draw both plots on grid paper using the 40 cm string loop (1 cm = 1 m): one with width 8 cm and length 12 cm, the other with width 12 cm and length 8 cm, and turn one around to see that they are the same rectangle. They compare with 4 m by 24 m (area 96 m ² but needs 56 m of wire) to see why only x and $20 - x$ pair up with a fixed fence. They check the context rule $0 < x < 20$ for both roots. They also link back to the area table (rows $x = 8$ and $x = 12$, both 96) and note that the roots add up to 20.	Say: 'Chebet and Kiprop said two different plots give the same area. Use today's mathematics to settle it.' Give groups 5 minutes with the string loops and grid paper. Circulate and ask: 'What is the length when the width is 8? When the width is 12?' and 'What do you notice about the two roots and the number 20?' WAIT TIME 10 seconds after each question. Then hold a whole-class talk: cold-call 3 groups in turn: 'Is it one plot or two plots?' and press: 'Say it in terms of width and length.' Make sure the language is plain: 'The 8 m by 12 m plot turned round is the same rectangle.' Ask: 'Why do the roots add up to 20?' WAIT TIME 15 seconds, then cold-call 2 learners (width + length = 20, so if x is a width the other root is the length). Show 4 m by 24 m on the board: 'It also has area 96 m ² , so why is it not on Mama Wanjiru's list?' (Cold-call 2: it needs 56 m of wire, not 40 m.) Finally ask: 'Should we accept every root?' and have learners confirm 8 and 12 both lie between 0 and 20.	Model-based explanation with physical turning of the string-loop rectangle: manipulating the concrete model resolves the conflict between 'two answers' and 'one plot', and connects the abstract roots to the symmetry seen in the area table. Group talk using sentence frames ('The roots are 8 and 12 because ... This means the plot ...') supports precise mathematical language.	Use a 1-minute exit explanation on a slip: 'Explain in two sentences why $x = 8$ and $x = 12$ are not two different plots.' Success indicator: the learner states that the 8 m by 12 m plot is the same rectangle with width and length swapped, and that width + length = 20 m. Look for the misconception that 'two answers means two different shapes'. At least 75% of slips should include the 'turned round' idea; re-address with the string loop for those who do not.
Driving Question Board (DQB) Creation	Learners return to the class Driving Question Board. In pairs they sort the existing question cards into 'now answered' and 'still	Say: 'Look at our board. Which questions can we cross off or answer now, and what is your evidence?' Cold-call 3 learners to	Driving Question Board: making the growth in understanding visible lets learners track what evidence resolved a question and turns	Check the new question cards. Strong questions extend the phenomenon (e.g. what happens at $x = 10$, what if the target is

 VIDEO: Zero product property Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Solving equations graphically: word problems <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	open'. They answer the question 'Why do two different widths give the same area?' with an evidence card citing the roots 8 and 12 and the string-loop rectangle. Each learner writes at least one new question on a card, for example: 'What if the target area is 100 m²?', 'What if the target is 105 m²?', 'Does every quadratic equation have two roots?', 'Why is the maximum 100 m²?', 'What if the equation will not factorise?'. Learners place cards on the board under headings: Solving, Context checks, Maximum area, Other situations.	move a question card and read the evidence aloud. WAIT TIME 10 seconds before each cold-call. Say: 'Now write one new question that today's results make you wonder about.' Give 2 minutes. Cold-call 3 learners to read new questions and pose a follow-up: 'Which lesson do you think will help answer this?' Point out that the question about 100 m² and 105 m² will come back in later lessons and that other situations such as a mandazi tray and a thrown ball come next. Do not answer these new questions yet; record them as 'to be investigated'.	remaining uncertainty into purposeful inquiry for the next lessons.	above 100) rather than repeat what was learned. Success indicator: at least 80% of learners write one new question that links to the fence, the area or the number of roots, and 3 or more pairs can cite specific evidence (the factorised equation or the string-loop rectangle) for a card they moved.
Model Building Phase  VIDEO: Solving quadratic equations: complex roots <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Quadratics by taking square roots: with steps <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners revise the class cumulative model of Mama Wanjiru's plot. The model currently holds the area models (Lesson 2) and all three identities (Lesson 3). In groups of 4 they add a new panel titled 'Solving for a target area': (1) $A = x(20 - x) = 20x - x^2$ (from Lesson 1), (2) target equation $x(20 - x) = 96$ becoming $x^2 - 20x + 96 = 0$, (3) factorised form $(x - 8)(x - 12) = 0$, (4) zero-product property giving $x = 8$ or $x = 12$, (5) a sketch showing the 8 m by 12 m rectangle and the same rectangle turned round, and (6) a note: width + length = 20, so the two solutions are x and $20 - x$. They mark 'Model v2: two solutions'. Learners add an arrow labelled 'Still to explain: maximum area' and one 'check against the context $0 < x < 20$' tick box.	Say: 'Our model must now explain why two widths give 96 m². What new part must we add?' WAIT TIME 15 seconds. Cold-call 3 groups for a proposal, then agree on the panel. Circulate and ask each group: 'Show me where the model says why there are two answers.' and 'Where does the 20 m come in?' Ask one group to present and cold-call 2 other groups to challenge: 'What is missing or unclear?' Ask: 'What would the model predict for a target of 84 m²?' (Cold-call 2: $x^2 - 20x + 84 = 0$ gives 6 and 14; do not solve fully, as this is Lesson 8 content, just have them predict that two widths appear and add to 20.) Close with: 'Which of your predictions at the start was right, and what changed in your thinking?' Cold-call 3 learners.	Cumulative model revision: learners integrate new evidence into a shared representation, test it with a new prediction and identify gaps. This makes explicit the link between algebra (roots), geometry (rectangle turned round) and context (fence length), and reveals what the model still cannot explain (maximum area).	Check each group's panel for: correct equation chain, both roots, the 'turned round' sketch, and the note that the roots add to 20. Success indicators: all groups include both solutions and link them to the same rectangle; at least 3 groups can predict that a different target area would again give two widths summing to 20. Collect an exit ticket: 'Solve $x^2 - 20x + 96 = 0$ and say what the answers mean for Mama Wanjiru.' At least 80% should score full marks.

D. TEACHER REFLECTION

After the lesson, ask: Did learners see why we rearrange to zero before using the zero-product property, or did they treat it as a rule? Which learners still made sign errors when factorising $x^2 - 20x + 96$, and do they need a Lesson 4 refresher with tiles? How many learners explained the two roots as the same rectangle turned round rather than two shapes? Which new DQB questions most strongly point to Lessons 7 and 8 (other situations, maximum area, impossible targets)? What would you change about the wait times or cold-call counts so quieter learners participate?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We saw that $x(20 - x) = 96$ can be rearranged to $x^2 - 20x + 96 = 0$ and factorised as $(x - 8)(x - 12) = 0$. Substituting $x = 8$ and $x = 12$ into $x(20 - x)$ both gave 96 m ² , while $x = 9$ did not make the product zero.
What did I learn?	We learned that a quadratic equation is solved by rearranging to zero, factorising and using the zero-product property: if $p \times q = 0$ then $p = 0$ or $q = 0$. We also learned to check each solution against the context $0 < x < 20$.
How does this explain the phenomenon?	The two solutions $x = 8$ and $x = 12$ appear because width + length = 20 m, so if the width is 8 m the length is 12 m and vice versa. They describe the same 8 m by 12 m plot turned round, not two different shapes, which resolves the puzzle in Mama Wanjiru's plot. Our model now includes two solutions that add up to 20.

LESSON 7 (40 minutes (one lesson; double lesson recommended if the ball activity is done outdoors)): Forming and Solving Quadratic Equations from Real-Life Situations: Mandazi Tray, Path Around the Shamba and Thrown Ball

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners translate three worded situations (mandazi tray, path around a shamba, thrown ball) into quadratic equations by defining a variable, writing the equation and rearranging to zero. They solve by factorisation and decide which solutions fit the context. This extends the driving question: the same steps that found the 8 m and 12 m widths for Mama Wanjiru's plot work for many real situations.
Knowledge	By the end of the lesson the learner should be able to form quadratic equations from real-life situations (KICD sub-strand outcome: form, solve and apply quadratic equations in real-life situations). Learners will know that: (1) a word problem becomes an equation once the unknown is defined with a letter and units; (2) the equation is rearranged to the standard form $ax^2 + bx + c = 0$ before factorising; (3) the zero-product property gives up to two roots; (4) each root must be checked against the context, so a negative length or width is rejected, while in the ball situation both $t = 1$ s and $t = 3$ s are valid.
Skills	NGSS-aligned practices: Developing and using models (draw the tray, path and ball path to represent the situation); Using mathematics and computational thinking (form, rearrange, factorise and solve); Analysing and interpreting data (check roots against the context and an area or height table); Constructing explanations (justify why a root is kept or rejected); Engaging in argument from evidence (defend choices in a gallery walk). Core competencies: Critical thinking and problem solving; Communication and collaboration.
Attitudes	Integrity: checking every answer honestly, even when it looks neat. Responsibility: using the fence-wire, grid paper and string models with care. Unity: sharing roles in groups. Appreciation of how mathematics supports decisions for farmers, traders and athletes in Kenya. PCI link: Life skills (decision making) as learners decide which answers make sense.
Key Inquiry Question	How do we form quadratic expressions and equations? Where do we apply quadratic equations in real-life situations? (KICD key inquiry questions; wording to be verified against the KICD page.) Lesson question: How do we turn a real situation into a quadratic equation, and how do we know which solutions make sense?
Purpose in Storyline	Lesson 6 resolved the central puzzle: $x(20 - x) = 96$ gives $x^2 - 20x + 96 = 0$, $(x - 8)(x - 12) = 0$, so $x = 8$ or 12 , the same 8 m by 12 m plot turned round. Lesson 7 shows that this is a general method by giving learners three new situations to form and solve (steps from L1 forming expressions from areas and L4 factorising). Learners meet two outcomes: a root that must be rejected (negative length) and a situation where both roots are valid (the ball at 15 m). This prepares Lesson 8, where they design a plot and explain why 105 m ² is impossible. The DQB and the cumulative model gain a section on checking roots against the context.
Safety Notes	Ball activity: do it outdoors on open ground at least 5 m from windows, people and roads, or use a paper ball or bean bag thrown gently upwards in the classroom with no one under it. Keep walkways clear of bags. Scissors for cutting grid paper: blunt-ended, used seated. Strings and loops kept off necks and not pulled tight. If no outdoor space exists, use the teacher demonstration plus the table of h-values only.

B. LESSON OVERVIEW

Setting: Mama Wanjiru in Kericho now wants to use her mathematics beyond the vegetable plot. Her children Chebet and Kiprop tell her that the same steps she used for the 96 m² plot (write the area with x , set it equal to the target, rearrange to zero, factorise, check) can solve other questions. In this lesson learners work in groups of 4 through three situations: (a) a mandazi tray at a Kisumu market stall holding 60 mandazi in x rows and $(x + 4)$ columns; (b) a path of uniform width w m around a 12 m by 8 m shamba in the Rift Valley, where shamba plus path covers 140 m²; (c) a football kicked at Kasarani whose height is $h = 20t - 5t^2$ metres after t seconds (g taken

as 10 m/s^2). Learners predict first, then form equations, solve by factorisation, and test every root against the situation. They find $x = 6$ (–10 rejected), $w = 1$ (–11 rejected), and $t = 1$ and $t = 3$ both valid for $h = 15$, and compare with the plot, where $x = 8$ and $x = 12$ were both valid but widths above 20 m were impossible. The phenomenon link: each equation arises from a fixed total (60 mandazi, 140 m^2 , 15 m height) just as the fixed 40 m wire and 96 m^2 area did. The lesson ends with a DQB update and revision of the cumulative model to add the step 'check roots against the context'.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Quadratic equations word problem: triangle dimensions Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Zero product property Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners recall the 40 m fence puzzle ($x(20 - x) = 96$ gives $x = 8$ and $x = 12$) and work individually for 3 minutes on a prediction card. Card: 'A Kisumu trader arranges 60 mandazi in a rectangle with x rows and 4 more columns than rows, so x rows and $(x + 4)$ columns. Without calculating, predict: how many rows could there be? Could the answer be a negative number of rows? Why or why not?' Learners then predict about the ball: 'A ball kicked upwards reaches the height of 15 m. How many times do you think it passes the height 15 m, and when?' Learners record predictions with reasons in their books, then share with a partner (2 minutes).	Display the phenomenon picture of Mama Wanjiru's plot and the row of areas: '8 m gives 96 m^2 , 12 m gives 96 m^2 .' Say: 'Last lesson we solved $x^2 - 20x + 96 = 0$. Today we ask: how do we start from a story and get to an equation like that?' Read the mandazi situation aloud and draw a 3 by 5 tray on the board to show rows and columns. Say: 'Write your prediction quietly. You have three minutes. No talking yet.' WAIT TIME: after asking 'How many times does the ball pass 15 m?', wait 12 seconds in silence before taking answers. Then cold-call 4 learners (not volunteers) for predictions on the number of rows and 3 learners for the ball. Record every prediction on the board WITHOUT saying which is right. Ask 2 learners: 'What made you think so?' Circulate and note the most common ideas, such as 'the answer is just 60 divided by 4' or 'negative is impossible'.	Predict-Observe-Explain (the Predict step): committing to a prediction in writing makes learners aware of what they currently believe, so later evidence can be compared with it. Recording all predictions publicly, without judging them, builds a class record of ideas to test and shows that mistakes are useful.	Look for: (1) learners who can name rows and columns as a product $x(x + 4)$ in words or a sketch; (2) learners who treat negative as impossible for rows, which shows context sense; (3) learners who think the ball passes 15 m only once. Tick a checklist (names versus 'product idea', 'context sense', 'ball once or twice'). If more than a third of learners cannot link rows and columns to multiplication, revisit the 3 by 5 tray before moving on.
Observe Phase  VIDEO: Quadratic equations word problem: box dimensions Source: Khan Academy (English - US curriculum)	Stage A (about 7 min): Mandazi tray. Groups draw the tray with x rows and $(x + 4)$ columns using squared paper (a 6 by 10 tray of squares is cut out to test). They write the equation $x(x + 4) = 60$, expand to $x^2 + 4x = 60$ and rearrange to $x^2 + 4x - 60 = 0$. Stage B (about 8 min): Path	Give each group squared paper, a ruler, scissors (blunt) and a recording sheet with three columns: 'Define the variable', 'Write the equation', 'Rearrange to zero'. Say: 'Step one is always the same: say what your letter means, with units.' Model once at the board: 'Let x be the number of	Modelling (Developing and using models): drawing the tray, path and ball path turns words into a picture and then an equation, so learners can see where each term comes from. Structured three-step recording (define, write, rearrange to zero) gives a repeatable routine, the same one used for $x(20 - x) = $	Observable indicators: (1) the variable is defined with units (e.g. ' w = width of path in metres'); (2) the outer dimensions of the path are written as $12 + 2w$ and $8 + 2w$ and not $12 + w$; (3) the equation is rearranged to zero with correct signs ($x^2 + 4x - 60 = 0$; $w^2 + 10w - 11 = 0$); (4) the ball table shows $h $

Search ARES for similar videos EXERCISE: Zero product property Source: Khan Academy (English - US curriculum) Search ARES for similar readings	around the shamba. Groups draw a 12 m by 8 m rectangle (scale 1 cm = 1 m) and a path of uniform width w on all sides. They label the outer rectangle $(12 + 2w)$ by $(8 + 2w)$ and write $(12 + 2w)(8 + 2w) = 140$. They expand: $96 + 24w + 16w + 4w^2 = 140$, which simplifies to $4w^2 + 40w - 44 = 0$, then dividing by 4 gives $w^2 + 10w - 11 = 0$. Stage C (about 8 min): Thrown ball. Groups complete a table for $h = 20t - 5t^2$ at $t = 0, 1, 2, 3, 4$ ($h = 0, 15, 20, 15, 0$) and sketch the path of the ball. They notice $h = 15$ appears twice ($t = 1$ and $t = 3$), like the repeated area 96 m^2 in the fence table at $x = 8$ and $x = 12$. For each stage they record the variable definition, the equation, and the equation rearranged to zero. Note: the factorising is done in the Explain phase.	rows.' Circulate during Stage A and ask: 'If there are x rows, how many columns? How do you know?' During Stage B ask: 'Why is the path added twice to the length, not once?' (a path lies on both sides). Point to the drawing and wait 10 seconds for the answer. For Stage B support, hand out the pre-drawn outer rectangle. During Stage C say: 'Look at your table. Which height appears twice? What does that remind you of in our fence table?' Cold-call 3 groups to report the table and the repeated height. Deliberately do not factorise for groups yet. Check each group has written all three steps for each situation. Early finishers: write what $x = 5$ or $x = 7$ would give in the mandazi tray ($x(x + 4) = 45$ or 77), to confirm that only $x = 6$ gives 60.	96. Comparing the repeated heights in the ball table with the repeated areas in the fence table builds the link that two answers can come from one equation.	$= 15$ at $t = 1$ and $t = 3$. Use the recording sheets as a quick check. Common errors to note: writing $(12 + w)(8 + w)$ for the path, forgetting to divide by 4, leaving the equation as $x^2 + 4x = 60$ and trying to factorise. Offer prompts rather than answers: 'Where is the path on each side?'
Explain Phase VIDEO: Quadratic equations word problem: triangle dimensions - dubbed(KY) Source: Khan Academy (Kiswahili) Search ARES for similar videos EXERCISE: Equations & inequalities word problems <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar	Groups factorise each equation and use the zero-product property from Lesson 6. Mandazi: $x^2 + 4x - 60 = 0$; two numbers with product -60 and sum $+4$ are $+10$ and -6, so $(x + 10)(x - 6) = 0$, $x = -10$ or $x = 6$. Path: $w^2 + 10w - 11 = 0$; numbers with product -11 and sum $+10$ are $+11$ and -1, so $(w + 11)(w - 1) = 0$, $w = -11$ or $w = 1$. Ball: $20t - 5t^2 = 15$ gives $5t^2 - 20t + 15 = 0$; dividing by 5 gives $t^2 - 4t + 3 = 0$, $(t - 1)(t - 3) = 0$, $t = 1$ or $t = 3$. Learners then complete a 'Keep or reject?' table: for each root they substitute into the original situation. The mandazi tray with $x = -10$ would mean -10 rows, so reject; $x = 6$ gives 6 rows of 10 mandazi ($6 \times 10 = 60$). The path with $w = -11$ would be a negative width, so reject; $w = 1$ gives an outer size of 14 m by 10 m = 140	Say: 'Use the sum-and-product method from earlier lessons. What two numbers multiply to -60 and add to 4?' Allow groups 3 minutes. Then cold-call 3 groups (one per situation) to write their factorised equations on the board while the rest check. Ask: 'Why do we get two answers each time? Which property tells us this?' WAIT 10 seconds, then take answers from 2 learners. Stress: 'Solving the equation is only half the job. Now we ask: does this answer make sense in the story?' Cold-call 4 learners: 'Can we have -10 rows of mandazi? Can a path have width -11 m?' For the ball ask: 'Is $t = 3$ an error because it is bigger than $t = 1$?' WAIT 12 seconds. Use a number line drawn on the board from 0 to 4 s with the ball path above it to show rising and falling.	Claim-Evidence-Reasoning: for each root learners state a claim (keep or reject), give evidence (substituting back into the story: $6 \times 10 = 60$, $14 \times 10 = 140$, $h = 15$ at $t = 1$ and 3), and reasoning (a length or number of rows cannot be negative; both times are valid because the ball goes up and comes down). Comparing three contexts alongside the fence puzzle builds the idea that the mathematics gives candidates and the context decides which ones are answers.	Indicators: (1) correct factorisations: $(x + 10)(x - 6)$, $(w + 11)(w - 1)$, $(t - 1)(t - 3)$; (2) correct use of the zero-product property, with each bracket set to zero separately; (3) every rejection includes a reason in context words, not just 'it is negative'; (4) learners explain why the ball keeps both roots. Use a 3-question exit slip: (i) Solve $x^2 + 2x - 15 = 0$ and state which roots make sense if x is a length in metres ($x = 3$; -5 rejected). (ii) Why does the ball have two valid times for 15 m? (iii) Why does Mama Wanjiru's plot have two valid widths? Sort slips into 'secure', 'partly' and 'needs support' piles for next lesson.

readings	m^2. For the ball, $t = 1$ s (rising) and $t = 3$ s (falling) both give 15 m, since $20 - 5 = 15$ and $60 - 45 = 15$, so both are kept. Learners write one sentence per situation beginning 'I kept / rejected ... because ...'. They then link to the phenomenon: for Mama Wanjiru's plot both $x = 8$ and $x = 12$ were valid, but a width of 25 m would be rejected because the length $20 - 25$ would be negative.	Challenge a group: 'Suppose someone says the ball reaches 25 m. Is that possible?' (set $20t - 5t^2 = 25$, which gives $t^2 - 4t + 5 = 0$ with no suitable factors; use the table, where the maximum is 20 m at $t = 2$). Then connect: 'In the ball and fence situations the table shows a maximum, 20 m for the ball and 100 m^2 for the plot. That is why some targets are impossible, and we will use this in Lesson 8.' Do not teach the maximum method here, only note it.		
Driving Question Board (DQB) Creation  VIDEO: Solving Real-World Problems Using Quadratic Equations: A Sample Application Source: CK-12  Search ARES for similar videos  EXERCISE: Quadratics by taking square roots: strategy <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Groups revisit the class DQB. Each group writes on sticky notes (a) one question from today that is now answered (e.g. 'How do we turn a story into an equation?' answered with the three steps: define, write, rearrange to zero; 'Why do some roots get rejected?' answered by context checks) and (b) one new question for the next lesson (e.g. 'How can we design a plot for a target area?' or 'Why can't a fence of 40 m give 105 m^2?' or 'Can any target between 0 and 100 m^2 be reached?'). They place answered notes in the 'We can explain' section and new notes under 'Still wondering'. The class reviews the original questions: 'Why do two plots have the same area?' (answered in L6 and revisited today), 'Why is there a maximum?' (partly answered, needs L8), and 'How can we find widths without trial and error?' (now answered for several situations).	Say: 'Take two minutes. Look at the board and our work. Write one question we can now answer and one we still wonder about.' WAIT 10 seconds before groups begin writing so everyone thinks. Cold-call 3 groups to read and place a note, and ask the class: 'Do you agree it is answered? What evidence do we have?' For each note, ask for the evidence from today's work. Highlight 'Why is there a maximum?' with a star and say: 'We have seen maximum values in the table for the plot (100 m^2) and the ball (20 m). We will explain this fully in Lesson 8 with $A = 100 - (x - 10)^2$.' Do not explain it now. Praise precise questions and gently reword vague ones ('Why is there a limit to the area?').	Driving Question Board: tracking questions answered and unanswered makes growth in understanding visible and gives learners ownership of what comes next. Requiring evidence for 'answered' notes builds scientific habits of argument rather than accepting claims.	Observable indicators: (1) answered notes cite evidence from today's equations (e.g. '$x = 6$ because $6 \times 10 = 60$'); (2) new questions point to the maximum, target areas or impossible plots; (3) at least 3 groups raise a question related to the limit on area. Record which groups have only recall-level questions and prompt them with 'What would happen if the target were bigger?'
Model Building Phase  VIDEO: Quadratic	Learners revise the cumulative class model, which now has: v1 area models (L2), the three identities (L3), and two solutions (L6). New for Lesson 7: a 'Form	Say: 'Our model must now do more than explain the 8 m and 12 m plots. It must show how to start from any story.' Display the blank flowchart and cold-call 5 learners	Cumulative modelling: adding a procedure (the flowchart) and worked cases to the existing model shows how each lesson extends the previous one and	Indicators: (1) the flowchart steps are in the right order, with 'rearrange to zero' before 'factorise' and 'check in context' after solving; (2) the paragraph

equations word problem: triangle dimensions Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Solving equations graphically: word problems <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	and solve' flowchart: Read the story, define the variable with units, write the equation, rearrange to $ax^2 + bx + c = 0$, factorise, use the zero-product property, check each root in the story, state the answer in a sentence. Learners add three worked examples around the plot model as evidence: mandazi $x = 6$ (–10 rejected), path $w = 1$ (–11 rejected), ball $t = 1$ and 3 (both kept). Each group also adds an arrow from the plot's valid range $0 < x < 20$ to the idea 'context sets the allowed values'. Individually, learners write a short paragraph (3 to 4 sentences) titled 'How my model explains Mama Wanjiru's two equal-area plots and why I must check my answers', using the words quadratic equation, factorise, root and reject.	to supply one step each in order, with the class giving thumbs up or down. Ask: 'Where on our model does the check against the story go?' WAIT 12 seconds. Then ask: 'If $x = 25$ were a root for the plot, what would we say?' (reject, the length $20 - 25 = -5$ is negative). Walk round while learners write paragraphs and choose 2 to read aloud, asking the class to listen for the three required words. Collect the model sheets. Close with: 'Next lesson you will design your own plot for the school garden with a target area, and decide when no plot is possible.' Remind learners to bring their notebooks and grid paper.	connects to the phenomenon. Writing an explanation in own words consolidates understanding and lets the teacher see misconceptions, such as treating every root as an answer.	correctly states that the two equal areas come from the roots $x = 8$ and $x = 12$ of the same equation (the same 8 m by 12 m plot turned round) and that invalid roots are rejected; (3) the model includes all three situations with correct values ($x = 6$, $w = 1$, $t = 1$ and 3). Mark with a 3-level rubric: complete and accurate, mostly accurate, needs support. Plan a short reteach for learners who miss the context check.
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D. TEACHER REFLECTION

After the lesson consider: (1) Which learners wrote $(12 + w)(8 + w)$ for the path, and did the drawing with the path on both sides correct it? (2) Did learners see that rejecting a root is a mathematical decision based on the story, not a mistake in the algebra? Look at the justification sentences. (3) Did the ball example help learners accept that both roots can be valid? Which learners still thought one root had to be 'wrong'? (4) Was the time split between the three situations realistic? Consider giving the ball task as homework if time runs short. (5) Which groups struggled with factorising when the product was negative (–60, –11)? Plan extra sum-and-product practice. (6) What does the DQB show about readiness for Lesson 8, especially questions about the maximum area and impossible targets such as 105 m²? Note any misconceptions to address, for example that 'any area is possible if you choose x well'.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that a worded situation (60 mandazi in x rows and $(x + 4)$ columns; a path of width w around a 12 m by 8 m shamba with total area 140 m ² ; a ball with $h = 20t - 5t^2$) can be written as an equation and rearranged to zero. We saw that the ball reaches 15 m twice ($t = 1$ and $t = 3$), like the 96 m ² area appearing at $x = 8$ and $x = 12$ on the 40 m fence.
What did I learn?	We learned the steps: define the variable with units, write the equation, rearrange to $ax^2 + bx + c = 0$, factorise, use the zero-product property, then check each root in the situation. The mandazi equation $x^2 + 4x - 60 = 0$ gives $(x + 10)(x - 6) = 0$; the path equation $w^2 + 10w - 11 = 0$ gives $(w + 11)(w - 1) = 0$; the ball equation $t^2 - 4t + 3 = 0$ gives $(t - 1)(t - 3) = 0$.
How does this explain the phenomenon?	We can explain that a quadratic equation can have two roots but the context decides which ones are answers: $x = -10$ rows and $w = -11$ m are rejected because lengths cannot be negative, so the answers are $x = 6$ and $w = 1$, while $t = 1$ s and $t = 3$ s are both

	kept because the ball goes up and comes down. This is why Mama Wanjiru's plot has two valid widths ($x = 8$ and $x = 12$, the same 8 m by 12 m plot turned round) but widths above 20 m are impossible.
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LESSON 8 (80 minutes (2 x 40-minute lessons or one double lesson)): Applying and Synthesising: Design a School-Garden Plot and Explain the Maximum

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners combine forming, factorising and solving to design and justify a school-garden plot for a target area and a fixed fence length, and explain why the area at a fixed fence length has a maximum of 100 m ² at $x = 10$, completing the Final Explanation of Mama Wanjiru's plot in Kericho.
Knowledge	Learners will be able to: (1) form the area expression $x(20 - x) = 20x - x^2$ for a 40 m fence; (2) write a target-area equation in standard form $ax^2 + bx + c = 0$ and factorise it, e.g. $x^2 - 20x + 84 = 0 = (x - 6)(x - 14)$; (3) use the zero-product property to solve it and check both solutions against $0 < x < 20$; (4) use $A = 100 - (x - 10)^2$ to explain why no area above 100 m ² is possible, so a target of 105 m ² is impossible; (5) explain why the widths x and $20 - x$ give the same rectangle turned round.
Skills	Mathematical modelling, forming equations, factorising, justifying and arguing from evidence, constructing explanations, revising a model, communicating reasoning orally and in writing.
Attitudes	Perseverance and integrity in checking solutions; responsibility and respect when critiquing peers' designs; appreciation of quadratic equations in real-life planning such as gardens, trays, paths and thrown balls. Core competencies: Critical thinking and problem solving; Communication and collaboration. Values: Responsibility, Integrity, Unity. PCIs: Education for Sustainable Development (school garden and food security); Life skills (decision making).
Key Inquiry Question	How do we form quadratic expressions and equations? Where do we apply quadratic equations in real-life situations?
Purpose in Storyline	This is the final lesson of the storyline. Lessons 1 to 7 gave us the area expression (L1), the squares identities (L2), the difference of two squares (L3), factorisation (L4, L5), solving by factorisation with two solutions (L6) and forming equations from situations and rejecting invalid answers (L7). Lesson 8 uses all of these in a new design task, finishes the Driving Question Board (DQB) and completes the cumulative model so learners can answer the driving question: how can we find the dimensions of a plot when its fence length and area are both fixed, and why do two different plots sometimes have the same area?
Safety Notes	Use blunt-ended scissors and keep string loops, tiles and grid paper off the floor to avoid tripping. If measuring a real garden outdoors, stay within the marked area and carry tape measures carefully. Do not dig or fence any real ground without teacher and school permission.

B. LESSON OVERVIEW

Mama Wanjiru's children, Chebet and Kipchop, found that with 40 m of wire an 8 m wide plot and a 12 m wide plot both enclose 96 m², while a 10 m wide plot gives 100 m² and widths above 20 m are impossible. In this final lesson learners act as designers for their own school garden. Each group is given a 40 m fence and a target area of 84 m². They form the equation $x(20 - x) = 84$, rearrange it to $x^2 - 20x + 84 = 0$, factorise it as $(x - 6)(x - 14) = 0$, solve it, and check that a 6 m by 14 m plot uses exactly 40 m of wire. They then test a second target of 105 m² and explain why it is impossible using $A = 100 - (x - 10)^2$, which is an identity check using Lessons 2 and 3 and not a new method. Learners complete the full area table, finalise the DQB and write the Final Explanation, comparing their Lesson 1 model with the final model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Zero product property Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Zero product property Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners recall Mama Wanjiru's 40 m fence in Kericho. The teacher poses the school-garden challenge: 'The school wants a rectangular vegetable plot (sukuma wiki and spinach) using exactly 40 m of fencing wire. Could we get 84 m ² ? Could we get 105 m ² ?' Individually, learners write a prediction in their books for each target: possible or impossible? If possible, what widths might work? They give a reason in one sentence and note a confidence level from 1 to 5. Then they share with a partner.	Display the challenge on the board and say: 'Remember Chebet and Kipchop's plots. Without calculating, predict whether each target can be reached with 40 m of wire. Write your reasoning.' Give WAIT TIME of 15 seconds of silent thinking before writing. Circulate and note predictions without confirming. Cold-call 3 pairs for the 84 m ² target and 3 pairs for the 105 m ² target, and ask each, 'What from our earlier evidence makes you say that?' Record the claims on the board in two columns (possible / impossible) without judging. Say: 'We will test every claim using our full model.'	Predict-Observe-Explain (POE) with Claim-Evidence-Reasoning: learners commit to a claim before testing, so the final evidence either confirms or challenges their thinking and they see their own growth in understanding.	Look for: (1) predictions that mention the 100 m ² square at $x = 10$ or the table from Lesson 1; (2) use of the idea that length + width = 20; (3) misconceptions such as 'a longer fence always gives more area' or 'any target below 105 is possible'. Tally on a checklist how many learners predict 105 m ² impossible and how many give a reason.
Observe Phase  VIDEO: Solving quadratics by factoring Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Zero product property Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	In groups of 4, learners design the plot. Step 1: Define the variable: width = x m, length = $(20 - x)$ m because $2(\text{length} + \text{width}) = 40$. Step 2: Form the equation $x(20 - x) = 84$. Step 3: Expand and rearrange to standard form: $20x - x^2 = 84$, so $x^2 - 20x + 84 = 0$. Step 4: Factorise using the sum-and-product method: two numbers with sum -20 and product 84 are -6 and -14, so $(x - 6)(x - 14) = 0$. Step 5: Use the zero-product property: $x = 6$ or $x = 14$. Step 6: Check against $0 < x < 20$: both are valid. A plot 6 m wide has length 14 m, and a plot 14 m wide has length 6 m, so it is the same 6 m by 14 m rectangle turned round. Step 7: Verify: $6 \times 14 = 84$ m ² and $2(6 + 14) = 40$ m. Groups build the plot with a 40 cm string loop (1 cm = 1 m) on grid paper. Step 8: Test the second target, 105 m ² : $x^2 - 20x + 105 = 0$. Learners search for two numbers with sum -20 and product 105 and find none that	Give each group a design card, a 40 cm string loop, grid paper, area tiles and the blank area table. Say: 'You are the garden planning committee. Show every step: variable, equation, standard form, factorise, solve, check.' Circulate and ask: 'How do you know the length is $20 - x$?' and 'Why must we write the equation equal to zero before factorising?' For groups stuck on factor pairs, prompt: 'List the pairs of numbers that multiply to 84. Which pair adds to 20?' Give WAIT TIME of 10 seconds after each question before offering hints. For the 105 m ² target, ask: 'What do you notice when you look for numbers that multiply to 105 and add to 20?' Do not tell them the answer yet. Ask each group to point to the largest value in their completed table and say what it tells them. Cold-call 4 groups to report the design result (width, length, check) and 2 groups to report what happened with 105 m ² .	Modelling with mathematics and structured procedure: learners apply the full cycle (define, form, rearrange, factorise, solve, check) to a new situation, which shows that the same method resolves the original phenomenon and new designs. The area table gives numerical evidence that links to the algebra.	Observable indicators: (1) the group writes the length as $(20 - x)$ before forming the equation; (2) the equation is rearranged correctly to $x^2 - 20x + 84 = 0$; (3) the factorisation $(x - 6)(x - 14)$ is checked by expansion; (4) both solutions are checked against $0 < x < 20$ and interpreted as one rectangle; (5) the learner reports that 105 m ² has no valid width and links it to the table maximum of 100. Use a checklist with a tick for each step. Common errors to note: forgetting to rearrange to zero, sign errors such as $(x + 6)(x + 14)$, or treating $x = 6$ and $x = 14$ as two different shapes.

	work with whole numbers. They then complete the full area table from $x = 2$ to $x = 18$ (2, 36; 4, 64; 5, 75; 6, 84; 8, 96; 10, 100; 12, 96; 14, 84; 15, 75; 16, 64; 18, 36) and notice that the greatest value is 100.			
Explain Phase  VIDEO: Solving quadratics by factoring - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Number of solutions of quadratic equations <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners explain the maximum. The teacher writes $A = 20x - x^2$ on the board and asks whether it can equal $100 - (x - 10)^2$. Learners expand $(x - 10)^2 = x^2 - 20x + 100$ using the identity $(a - b)^2 = a^2 - 2ab + b^2$ from Lesson 2, so $100 - (x - 10)^2 = 100 - x^2 + 20x - 100 + 100 = 20x - x^2$. They carefully work: $100 - (x^2 - 20x + 100) = 100 - x^2 + 20x - 100 = 20x - x^2$. So $A = 100 - (x - 10)^2$. Pairs then answer: What is the smallest value $(x - 10)^2$ can take? (0, because a square cannot be negative.) So A is at most 100, reached when $x = 10$, a 10 m by 10 m square. Hence 105 m² is impossible and every width above 20 m gives a negative length. Learners also explain symmetry: $A = 100 - (x - 10)^2$ gives the same area for $x = 10 + d$ and $x = 10 - d$, e.g. 12 and 8 both give $100 - 4 = 96$, and 14 and 6 both give $100 - 16 = 84$. Learners write a Claim-Evidence-Reasoning paragraph answering: why do two widths give the same area, and why is there a maximum?	Say: 'Our table suggests a maximum of 100. Now let us prove it using what we know about squares.' Ask learners to expand $(x - 10)^2$ in pairs and cold-call 3 pairs for the result, with WAIT TIME of 10 seconds. Ask: 'Can $(x - 10)^2$ ever be negative? Why not?' Wait 15 seconds, then cold-call 3 learners. Ask: 'What is the biggest A can be?' Then challenge: 'Chebet says she can reach 105 m² if she is clever. What do you tell her?' Cold-call 2 learners. Show that $x = 10 + 2$ and $x = 10 - 2$ both give 96. Say: 'Be clear: the 8 m plot and the 12 m plot are the same rectangle turned round, not two shapes. A 4 m by 24 m plot also has 96 m² but needs 56 m of wire.' Do not teach completing the square or any formula; this is only an identity check using Lessons 2 and 3. Provide sentence starters: 'The area is greatest when... because...'	Claim-Evidence-Reasoning with an identity check: learners connect the numerical table (evidence) to the algebraic identity (reasoning) so the maximum and the symmetry are explained by structure and not only by pattern.	Look for: (1) correct expansion of $(x - 10)^2$ with the middle term $-20x$; (2) the statement that a square is never less than 0; (3) a conclusion that $A \leq 100$ and equality only when $x = 10$; (4) a clear statement that x and $20 - x$ give the same rectangle; (5) rejection of 105 m² using the maximum. Score each CER paragraph with a 3-point rubric: claim stated, evidence from the table or equation, reasoning from the identity. Follow up with learners who say the 8 m and 12 m plots are different shapes.
Driving Question Board (DQB) Creation  VIDEO: Zero product property Source: Khan Academy (English - US curriculum)  Search ARES	The class returns to the DQB built across the unit. In groups, learners sort the question cards into: answered, partly answered and still open. For each answered question, they attach evidence from a lesson (for example the area table from L1, the squares identities from L2 and L3, the factorisation $(x - 8)(x - 12) = 0$	Say: 'Take a card from the board. Do we have enough evidence to answer it? Show where.' Give each group 5 cards. After sorting, cold-call 5 groups to read one answered question with its evidence and 2 groups to read one open question. WAIT TIME of 10 seconds before each response. Ask: 'Which lesson gave us the	Driving Question Board review: learners track and publicly connect questions to evidence, which shows the class how their understanding developed and identifies remaining gaps.	Observe whether each answered card is linked to specific evidence and the correct lesson idea. Check that learners can say which questions remain open and why. A card sorted as 'answered' without evidence is a misconception flag. Record the number of cards correctly linked per group.

for similar videos  EXERCISE: Zero product property Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	from L4 and L6, the rejected answers from L7 and the 84 m² and 105 m² designs from L8). They then write one new question about where else quadratic equations are used (for example the mandazi tray at a Kisumu market, a path around a shamba in the Rift Valley, a ball kicked at Kasarani) and add it to a 'Next questions' area.	evidence?' and 'What would convince a sceptic?' Move any question not fully answered to 'partly answered' and note what is missing. Praise precise evidence and ask learners to cite lesson ideas by name (for example 'zero-product property').		
Model Building Phase  VIDEO: Solving quadratics by factoring Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Zero product property Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners complete the cumulative model. Each group takes their Lesson 1 model (area as width $\times$ length, $x(20 - x) = 20x - x^2$, first area table with 96 m² at 8 m and 12 m, 100 m² at 10 m) and compares it with the final model: area model of $x(20 - x)$; identities $(a + b)^2$, $(a - b)^2$, $a^2 - b^2$; factorisation of $x^2 - 20x + 96 = (x - 8)(x - 12)$; two solutions $x = 8$ and $x = 12$ as one rectangle turned round; checking against $0 < x < 20$; and the maximum from $A = 100 - (x - 10)^2$, $A \leq 100$. On a poster they draw both models side by side, listing what the Lesson 1 model could do and what the final model adds. Each learner then writes the Final Explanation individually: 'With 40 m of wire the length is $20 - x$, so the area is $x(20 - x)$. Setting it equal to a target area gives a quadratic equation that we factorise and solve. The two solutions x and $20 - x$ are the same rectangle turned round, so the area is symmetric about $x = 10$. The area is greatest, 100 m², at $x = 10$ because $A = 100 - (x - 10)^2$, and a square cannot be less than 0, so no area above 100 m² is possible.' Learners write it in their own words and include one worked design (6 m by 14 m for 84 m²).	Say: 'Compare your first model with the model you have now. What can the new model do that the first could not?' Give WAIT TIME of 15 seconds, then cold-call 4 groups to share one added feature. Ask: 'Which lesson gave you that feature?' Display the driving question and say: 'Use your model to answer both parts: how do we find the dimensions, and why do two plots have the same area?' Circulate during the Final Explanation writing and use the success criteria: variable defined, equation formed, factorised, both solutions checked, symmetry explained, maximum explained with the identity. Invite 3 volunteers to read their explanations. Close with: 'Where else in Kenya could you use this? Think of a mandazi tray in Kisumu or a ball kicked at Kasarani.' Collect the explanations for assessment.	Cumulative model revision with Final Explanation: comparing the first and final models makes the growth of understanding visible, and writing a coherent explanation consolidates the whole storyline into one answer to the driving question.	Use a 4-point rubric for the Final Explanation: (1) forms $x(20 - x)$ from the 40 m fence; (2) forms, factorises and solves the target-area equation with a check; (3) explains the pair x and $20 - x$ as one rectangle turned round; (4) explains the maximum using $A = 100 - (x - 10)^2$. Model comparison: learners name at least 3 added features (the x^2 term, the identities, factorisation, two solutions, rejecting invalid ones, the maximum). Learners scoring below 3 receive a follow-up with the area table and tiles in the next lesson.

D. TEACHER REFLECTION

After the lesson, consider: (1) How many groups completed all six steps of the design cycle (define, form, rearrange, factorise, solve, check) without prompting? (2) Did learners explain 8 m and 12 m as one rectangle turned round and not two shapes? (3) Could learners use $(x - 10)^2 \geq 0$ to justify the maximum, or did they rely only on the table? (4) Which factorisation errors (sign errors, forgetting to rearrange to zero) remained? (5) How well did the DQB cards link to evidence from specific lessons? (6) Which learners need more support with forming equations from worded situations, and what resource (tiles, string loops, grid paper) helped them most? Note adjustments for the next unit, such as more practice with factor pairs or more time for the Final Explanation.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that with 40 m of wire the length is $20 - x$, so the area is $x(20 - x)$. A target of 84 m^2 gives $x^2 - 20x + 84 = 0$, which gives a 6 m by 14 m plot. A target of 105 m^2 has no valid width, and the area table never goes above 100 m^2 , reached at $x = 10$.
What did I learn?	We learned how to form a quadratic equation from a target area, rearrange it to zero, factorise it, use the zero-product property, and check the solutions against $0 < x < 20$. We learned that $A = 100 - (x - 10)^2$ shows the area can never exceed 100 m^2 .
How does this explain the phenomenon?	Two widths such as 8 m and 12 m, or 6 m and 14 m, give the same area because x and $20 - x$ describe the same rectangle turned round, so the area is symmetric about $x = 10$. There is a maximum of 100 m^2 at $x = 10$ because a square $(x - 10)^2$ cannot be less than 0, so no plot with 40 m of wire can enclose more than 100 m^2 .

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.